Where does the tariff offset go? Credibility, dollar risk and the renminbi's anchor

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Abstract

The renminbi fell against the dollar in the 2018–19 U.S.–China trade war but rose 4.29 percent in 2025, when tariffs climbed further. Using 97 tariff announcements, daily data for 2017–2026 and a calibrated model of credibility and strategic anchoring, we ask where the exchange-rate offset went. Markets priced about one fifth of the permanent offset in both wars, less for larger announcements, and forward premia show no deferred depreciation. After the April 2025 reciprocal tariffs the dollar fell against 17 of 19 currencies, safe havens or not, as U.S.-specific risk predicts. China tightened its dollar anchor through the central parity: the renminbi's sensitivity to the dollar fell from 0.508 in 2023 to 0.185 in 2026, shifting the dollar's 2025 fall into its effective rate. Overall, the 2018–19 bilateral response is not a stable elasticity: it depends on the dollar and on how tightly China anchors to its trade opponent's currency.

Keywords: Tariffs; Exchange rates; Renminbi; Central parity; Policy credibility; Dollar risk
JEL classification: F13; F31; F33; F41

1 Introduction

The two U.S.–China trade wars produced strikingly different exchange-rate responses. The textbook response to a tariff is an appreciation of the tariff-imposing country's currency, which offsets part of the tariff in import prices. Over 2018–19 the renminbi depreciated by 6.77 percent against the dollar. In 2025, as the United States raised its tariffs on Chinese goods and announced reciprocal tariffs on almost all of its trading partners, the renminbi appreciated by 4.29 percent against the dollar, although the benchmark of Jeanne and Son (2024) (JS) predicts a 2.95 percent depreciation for a floating renminbi, and the dollar depreciated by 7.24 percent against a fixed basket of floating currencies instead of appreciating by 2.40 percent (Appendix Table A3). Yet the renminbi's effective exchange rate depreciated by 0.64 percent over the same year (Figure 1). This paper asks where the exchange-rate offset to U.S. tariffs went, and why.

The question arises because China manages the renminbi against a basket dominated by the dollar, the currency of its tariff-imposing trading partner. The strength of this dollar anchor decides how a dollar movement is divided between the bilateral rate, which governs the incidence of a bilateral tariff, and the effective rate against all trading partners. A depreciation against the dollar is also a depreciation against the trade opponent, which the United States has treated as a policy matter since the currency-manipulator designation of August 2019 (U.S. Department of the

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Treasury, 2019) and the exchange-rate chapter of the 2020 Phase One agreement (Office of the United States Trade Representative, 2020).

Several explanations could account for the 2025 pattern. Markets may have doubted that the announced tariffs would last. The dollar may have fallen for reasons unrelated to trade, if the tariffs raised the risk of holding dollar assets. China may have tightened its dollar anchor, so that less of the dollar's fall reached the bilateral rate and more reached the effective rate. The smaller spot response may have been postponed into forward rates. And because the 2025 announcements were far larger than those of 2018–19, a comparison of signed announcements would confound the size of the shocks with their transmission. Bilateral spot rates alone cannot distinguish among these explanations.

We examine them with a common decomposition and three sets of evidence. The data are New York noon quotations of 20 currencies from the Federal Reserve's H.10 release, the daily central parity published by the China Foreign Exchange Trade System (CFETS), offshore renminbi forwards and an inventory of 140 policy records, 97 of them with tariff news, over 2,253 common trading days from January 2017 to August 2026. We divide each renminbi movement into a component that follows the dollar's movement against a fixed basket, with a loading estimated on days without tariff news, and an own component, and we measure each announcement's exposure by its permanent JS prediction, so that announcements of very different size are compared in the same units.[1] First, we compare two-day own-component responses with the JS prediction, with the share of announced tariffs that was later implemented and with forward premia. Second, we compare the dollar's response across 19 currencies on the largest announcement days. Third, we estimate the renminbi's dollar loading year by year alongside those of 19 other currencies and trace its change to the rule that sets the central parity. We read the evidence through one model of three linked decisions: a government decides whether to sustain an announced tariff, foreign investors how much of their dollar assets to hedge against the policy risk it creates, and the Chinese authority how much of the resulting dollar movement to pass into the renminbi under a strategic cost of depreciating against the anchor issuer. A calibration on the JS benchmark, with one credibility parameter fitted to announcement-window prices, turns its restrictions into magnitudes.

The results yield three main conclusions. First, markets priced about one fifth of the permanent offset, and forward prices give no sign that the rest was deferred. The own-component coefficient on the JS prediction is 0.20 in 2018–19 and 0.18 in 2025–26, statistically indistinguishable across rounds. Within U.S. measures on China it is 0.53 on the smaller half of announcements and 0.13 on the larger half ($p = 0.024$), as a convex cost of implementation implies. The priced share of announced increases was close to the share in force at the end of 2019 and of 2025, a moment the calibration does not use. Forward premia from one month to one year did not respond detectably to tariff news.

Second, the dollar's response depended on the source of the risk. On China-specific escalations, such as those of May 2019 and October 2025, the dollar appreciated on average, and most against the currencies that weaken in global financial stress. After the reciprocal tariffs of 2 April 2025 it depreciated against 17 of 19 currencies, by 115 basis points on average and by similar amounts against the Mexican peso and the yen; safe-haven status does not explain the fall, and the dispersion around it lines up with the announced tariff rates. This is the structure that U.S.-specific risk produces in our hedging block, which enters through this restriction because hedge ratios are not observed.

[1] Signed labels of $+1$ for escalation and -1 for de-escalation enter only the signed-news regressions and the controls. Responses span the announcement day and the next common trading day of New York noon quotations, so they are daily responses rather than intraday surprises. Inference uses a restricted wild cluster bootstrap over contiguous news clusters, 13 in each round (Section 2.4).

Third, China tightened its dollar anchor through the central parity. The renminbi's loading on the dollar fell from 0.508 in 2023 to 0.185 in 2026, with a single break on 13 November 2024 and the largest fall among eight Asian currencies, while across 19 other currencies the change is unrelated to the reciprocal tariff rates. The parity, which in 2017–18 passed about two thirds of the previous day's market gap into the next day's parity, became almost unresponsive to it in 2024–26 and leaned against appreciation as well as depreciation. The timing is consistent with a role for the trade dispute, since the loading also fell after the manipulator designation of 2019. The evidence on motive is weaker. In 2025–26 the loading was lower on days of dollar appreciation (0.13 against 0.24), as a strategic cost of bilateral depreciation predicts, but the difference is not statistically significant ($p = 0.146$) and the loading fell on both sides, so the strategic weight accounts for the asymmetry and the timing of the tightening rather than for all of it.

Together these results locate the offset. In 2025 the dollar's fall, passed through the tighter anchor, contributed 3.54 percentage points of effective renminbi depreciation, and an own-component appreciation of 2.90 points, accrued almost entirely outside tariff-news windows, offset most of it. At the 2026 loading about half of a given dollar movement enters the effective rate, holding the own component fixed, against about a sixth at the 2023 loading. In the model, a strategic cost of depreciating against the anchor issuer can tighten the anchor even as the dollar's insurance value falls, and the tighter rule moves U.S.-specific dollar risk into the anchoring country's effective exchange rate.

Four limits frame these results. The second-round magnitudes depend on the export elasticity and trade shares through which the JS benchmark compresses the very large 2025 rates: across the alternatives of Appendix Table C7 the second-round coefficient ranges from 0.09 to 0.49, while the first-round results change little. The benchmark is linearised, and the April 2025 escalation took bilateral rates above 100 percent. Chinese retaliation moved the renminbi against its statutory effect, which credibility, as a scaling of the prediction, cannot produce. And the match between priced and implemented shares holds at the end of 2025: after the Supreme Court ruled in February 2026 that the International Emergency Economic Powers Act does not authorise tariffs, the implemented share fell to 0.12, against a priced share of 0.21.

These findings extend four literatures. First, Jeanne and Son (2024) establish the offset in a two-country New Keynesian economy, and pass-through of the 2018–19 tariffs to U.S. import prices was near complete (Amiti et al., 2019; Fajgelbaum et al., 2020; Cavallo et al., 2021). Trade-war news strengthened the dollar in the first round (Khalil and Strobel, 2024; Carlomagno and Albagli, 2022; Filippou et al., 2025) and weakened it in 2025 (Ostry et al., 2025; Kalemli-Özcan et al., 2026; Hartley and Rebucci, 2025; Suwanprasert, 2026), and Benguria and Saffie (2025) and Ignatenko et al. (2025) study the financial-market and trade effects of the reciprocal tariffs. We show where the offset appears in a managed partner currency: its own component was priced at the same share in two rounds in which the dollar moved in opposite directions, and the rounds differ in how the dollar's movement was divided between the bilateral and effective rates.

Second, tariff uncertainty has real effects (Handley and Limão, 2017; Caldara et al., 2020). Probabilities of policy change can be recovered from trade flows (Alessandria et al., 2024), agents first expected the trade war to end quickly (Alessandria et al., 2025), and markets learn about tariff persistence (Merendino and Monacelli, 2026). We link the size of an announcement to the probability that it lasts and compare the credibility priced in announcement windows with what was implemented.

Third, financial demand and hedging move exchange rates (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021; Jiang et al., 2021; Liao and Zhang, 2025; Shin et al., 2025; Bacchetta et al., 2026), global uncertainty strengthens the dollar (Albagli et al., 2024), and the asset prices of April 2025 carry information about the demand for dollars (Kekre and Lenel, 2026). Other work

characterises the monetary response to tariffs (Bergin and Corsetti, 2023; Monacelli, 2025; Auray et al., 2025) and the role of exchange-rate adjustment in sustaining trade cooperation (Auray et al., 2022). Our hedging block makes the sign of the dollar premium depend on the source of the risk, and our rule describes the anchoring authority rather than deriving its optimal policy.

Fourth, anchors are measured from co-movements (Frankel and Wei, 1994, 2007; Ilzetzki et al., 2019) and related to reserve-currency choice and geopolitics (Chinn et al., 2024); intervention and stabilisation are modelled by Fanelli and Straub (2021), Davis et al. (2023), Bergin et al. (2025) and Hassan et al. (2023); and the renminbi literature documents the two-pillar parity rule (Cheung et al., 2018; Jermann et al., 2022), the currency's regional influence (Fratzscher and Mehl, 2014; McCauley and Shu, 2019) and U.S. pressure on its adjustment (Weiss and Wichowsky, 2018). Closest to our setting, Hassan et al. (2026) find that a trade war that erodes dollar safety leads small economies to re-peg away from the dollar. For an economy whose trade opponent issues the anchor, we document the opposite response, a tighter anchor implemented through the central parity, and derive the threshold at which it occurs.

The rest of the paper is organised as follows. Section 2 describes the setting, the currency decomposition and the data. Section 3 sets out the model and its calibration. Sections 4–7 present the evidence on credibility, the dollar, the anchor and the location of adjustment. Section 8 concludes with policy implications.

2 Setting, data and measurement

2.1 Two tariff rounds and a managed renminbi

The first round ran from the Section 301 findings of March 2018 to the Phase One agreement of January 2020. The United States raised additional tariffs on about USD 360 billion of Chinese goods in five implemented tranches, and China retaliated on most of its imports from the United States (Appendix Table A1). The second round began with the first tariff threat of the incoming administration on 25 November 2024. It brought additional tariffs on China under the International Emergency Economic Powers Act from February 2025, the reciprocal tariffs on almost all partners announced on 2 April 2025, a bilateral escalation to rates above 100 percent in April, the Geneva truce of May 2025, a renewed threat in October 2025 and a further truce. Figure 1 shows the average additional tariffs that these measures implied. The average additional U.S. tariff on Chinese goods exceeded its 2018–19 peak, but much of the increase was reversed within weeks.

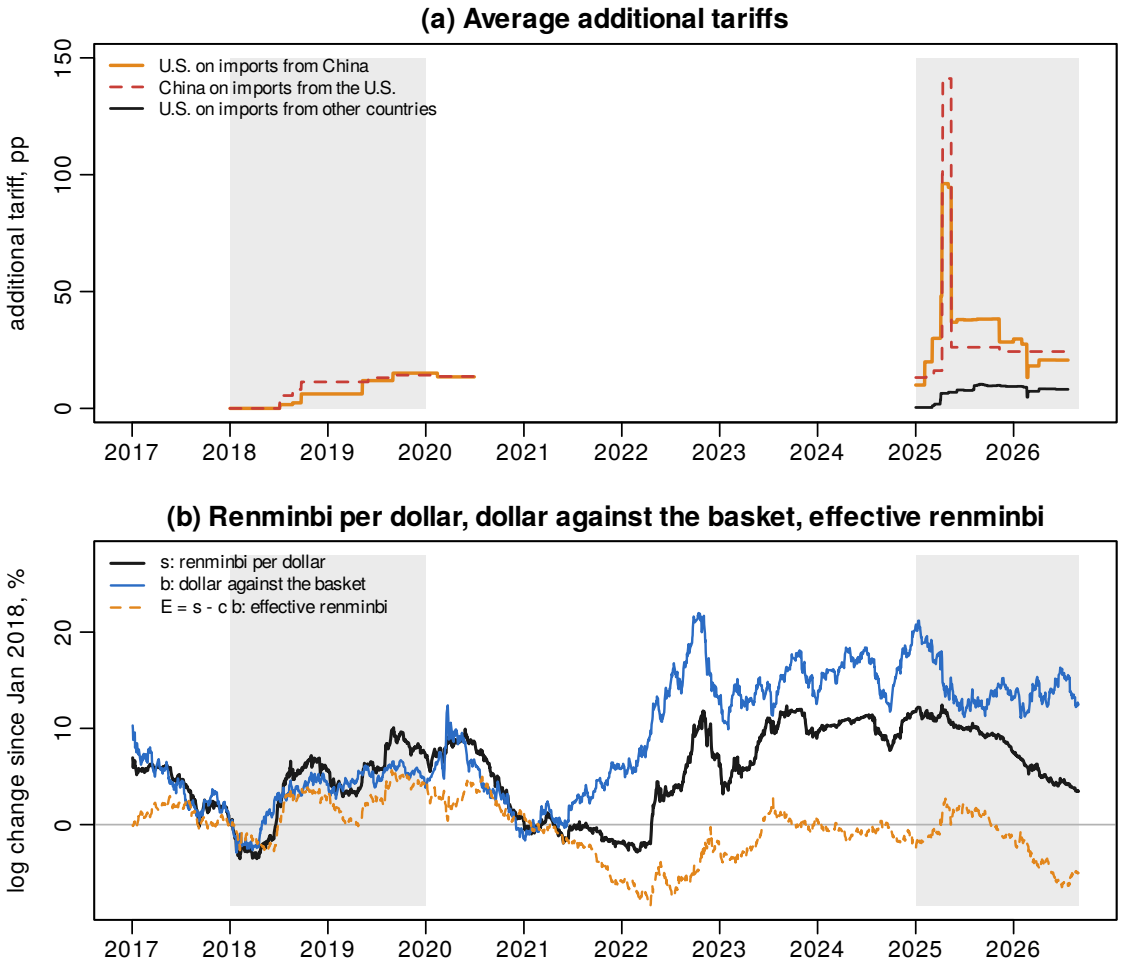
The renminbi is managed through a daily central parity, published by CFETS at 9:15 Beijing time, around which onshore trading is limited to a band of two percent (Jermann et al., 2022). Since August 2015 the parity has combined the previous day's closing rate with the overnight movement of a currency basket, the two pillars documented by Cheung et al. (2018) and Jermann et al. (2022), and from May 2017 a counter-cyclical adjustment (Jermann et al., 2022). The CFETS basket gave the dollar a weight of 22.4 percent in 2017 and 18.3 percent in its 2026 revision (Appendix Table A2).

2.2 A common currency decomposition

Let s denote the log renminbi price of a dollar and b the log value of the dollar against a fixed basket of floating currencies. An increase in s is renminbi depreciation against the dollar; an increase in b is dollar appreciation. With dollar-bloc weight w in the renminbi's effective basket, define $c = 1 - w$ and

$$E = s - cb, \qquad \Delta s = a\Delta b + x, \qquad \Delta E = (a - c)\Delta b + x. \tag{1}$$

Figure 1: Tariffs and exchange rates across the two rounds



Notes: Panel (a): import-weighted average additional tariffs, based on implemented tranches in the first round and the archived Budget Lab tracker in the second (9 June 2026 release, schedule through 23 July 2026). Panel (b): cumulative log changes since 2 January 2018 in the bilateral renminbi rate, the dollar against the fixed basket and the effective renminbi rate. Shading marks 2018–19 and 2025–26.

E rises with effective renminbi depreciation, a is the bilateral response to the dollar's common movement, and x is the renminbi's own component. A smaller a is a tighter dollar anchor. For an announcement window we write S, B and F for the changes in s, b and E, and χ for the own component.

The basket contains the 16 floating currencies of the 2017 CFETS basket quoted in the H.10 release, with their CFETS weights renormalised. The dollar bloc consists of the dollar, the Hong Kong dollar, the Saudi riyal and the UAE dirham, which gives $w = 0.320$ and $c = 0.680$. The weights are fixed throughout, so that changes in b reflect exchange-rate movements at constant weights. The New Taiwan dollar, the Indian rupee and the Brazilian real complete the 20 currencies used in the cross-currency comparisons.

2.3 Data and events

The H.10 noon quotations run from December 2016 to August 2026. A common trading day requires all basket quotations and a published central parity, which gives 2,253 days from 3 January 2017 to 31 August 2026. Treasury yields and the Chicago Board Options Exchange Volatility Index (VIX) come from their publishers, the second-round tariff schedule from the archived Budget Lab tracker, and trade values from the U.S. Census Bureau and China's customs administration. Onshore renminbi is denoted CNY and offshore renminbi CNH. CNH spot and forward rates at one, three, six and twelve months and the three-month dollar overnight index swap (OIS) rate are licensed Bloomberg closing quotes for 2017–19 and 2024–25. The replication package documents every source and the checks of each file against its publisher.

The event inventory records every official statement or legal instrument by the United States or China that changes, threatens to change or withdraws a threat to change tariffs on the other's goods, and U.S. tariff measures on third countries. Of 140 candidates, 119 are retained: 97 carry tariff news and 22 record implementation steps that convey no new policy. The 21 exclusions are investigations without a tariff decision, statements without a specified tariff, measures never officially announced and records whose direction or scope is unspecified. A signed bilateral label is $+1$ for escalation and -1 for de-escalation, with separate labels for U.S. measures on China, Chinese measures on the United States and U.S. measures on other partners. Each record is placed on the first common trading day on or after its release, and records on the same day are combined. The 2018–19 round contains 36 news days in 13 news clusters and the 2025–26 round 42 news days in 13 clusters (Appendix Table A4).

Announced exposure is measured from the announced tariff rate, covered imports and tariffs already in force. The tables and figures label this exposure as the "dose". The JS substitution mechanism converts these tariff changes into effective import and export tariffs, and the JS block gives the permanent prediction for the dollar component, B^{JS}, and for the own component, χ^{JS}. We retain the JS elasticities and first-round trade shares and use 2024 trade values for the second round (Table 1 and Appendix Table A3).

2.4 Estimation

The dollar loading is the coefficient in a regression of Δs on Δb on days without tariff news, in the tradition of Frankel and Wei (1994), estimated by year or by round with heteroskedasticity-robust (HC1) standard errors. The own component in a window subtracts the dollar change times the loading of the period in which the window begins; a second measure uses the loading estimated over the 250 preceding common days, which is known before the announcement. For an outcome

$y_t(h)$ cumulated from day t through $t+h$, the calibration regression is

$$y_t(h) = \alpha_r + \mu_r p_t + \delta_r \sum_{j=1}^{h} p_{t+j} + \eta_r \sum_{j=1}^{h} p_{t-j} + u_t(h), \tag{2}$$

with coefficients by period r. The prediction p_t is the announcement-day permanent JS prediction, which measures announced exposure, and the lead and lag sums separate that day's news from news on neighbouring days. The principal window covers the announcement day and the next common day ($h = 1$), so that a release after the New York noon quote is captured. A signed-news regression replaces p_t with the signed labels, and every model prediction passes through the same regression as the observed outcome.[2]

3 A model of credibility, dollar risk and the anchor

The evidence of Sections 4–7 is organised by three choices. A government decides whether to sustain an announced tariff change, foreign investors decide how much of their dollar assets to hedge, and the Chinese authority decides how much of the dollar's common movement passes into the renminbi. Each decision yields testable restrictions. The benchmark for every price response is the JS real block, which we retain unchanged. With no financial premium, an unanticipated permanent tariff moves the home currency by $de = -\kappa_m d\tau + \kappa_x d\tau^*$ and leaves the policy rate unchanged. Here e is the home-currency price of foreign currency, τ the effective import tariff and τ^* the effective tariff on home exports; at the JS calibration κ_m is 0.296 for the United States and 0.286 for China, and $\kappa_x = 1$. Appendix B sets out the linearised real block, the clearing of the currency markets and all proofs.

3.1 Credibility of a tariff announcement

An announcement proposes a change of absolute size $D > 0$ and sign $\varepsilon \in \{-1, 1\}$ in the tariff path. After the announcement the government learns the benefit v of sustaining the change; its continuation payoff is $vD - \delta D^2$ relative to reversal, where the quadratic term represents convex implementation costs. Continuation therefore occurs when $v \geq \delta D$, and its probability declines with D for any distribution of v. We take the survival function $\Pr(v \geq x) = 1/(1+x)$, which delivers a one-parameter credibility function. A reversed change lasts for an exponentially distributed interval with mean $1/\rho_s$; a sustained change remains. Under the bargaining interpretations of Ostry et al. (2025) and Alessandria et al. (2025), larger threats represent negotiating positions that are less likely to be implemented, consistent with declining credibility as announcement size increases.

Proposition 1 (Credibility, the expected exposure and policy risk)**.** *The probability that the announced change lasts is*

$$p(D) = \frac{1}{1+\delta D} = \frac{1}{1+D/D_0}, \qquad D_0 = \frac{1}{\delta}. \tag{3}$$

For an unanticipated announcement, the expected incremental tariff path, long-run mean and long-run variance are

$$n(u) = \varepsilon D[p(D) + (1-p(D))e^{-\rho_s u}], \tag{4}$$

$$m(D) = \frac{D}{1+\delta D}, \qquad V(D) = \frac{\delta D^3}{(1+\delta D)^2}. \tag{5}$$

[2] All announcement tests use one set of 4,999 six-point Webb draws (Webb, 2023) in a restricted wild cluster bootstrap (Cameron et al., 2008; MacKinnon and Webb, 2017; Roodman et al., 2019), seed 20261028, shared across equations. Days within five common days of a news day form contiguous news clusters; all other days are clustered by calendar month. Tables report cluster-robust (CR1) standard errors and bootstrap p-values.

For $\delta > 0$, $m'(D) > 0$, $m''(D) < 0$ and $V'(D) > 0$.

We measure D in basis points of the permanent JS prediction for the renminbi's own component (for news about other partners, the dollar prediction). The scale D_0 is then the announcement size whose credibility is one half, and $m(D)$ is the expected lasting response. A larger announcement raises expected long-run exposure at a decreasing rate and increases the variance of long-run policy, which links credibility to the portfolio risk of Section 3.2. At a given announced exposure, the own-component and H-maturity forward-premium responses are

$$X(D) = p(D)X_\infty(D) + (1 - p(D))X_s(D, \rho_s), \qquad P_H(D) = (1 - p(D))P_{H,s}(D, \rho_s). \tag{6}$$

The permanent tariff path moves the spot rate and leaves the forward premium unchanged; the temporary tariff path moves both. The spot response and one forward premium therefore identify (p, ρ_s) locally (Proposition B.1 in Appendix B.2), and as the temporary tariff path becomes shorter, the forward response vanishes and the spot response approaches $p(D)$ times the permanent prediction. The implemented share of announced tariffs provides a measure of credibility that is independent of announcement-window prices.

3.2 Dollar risk and hedging

Foreign investors hold a predetermined dollar position A and choose the unhedged amount n, with risk aversion ϱ and a convex cost of changing the hedge. Dealers absorb the residual position at a marginal cost Γ of bearing currency risk (Gabaix and Maggiori, 2015), so that private hedging and official transactions clear in the same market at a premium ℓ on dollar claims (Appendix B.3). A tariff announcement adds a policy innovation of variance $V(D)$, which changes the dollar's excess return by r_j and the payoff on the rest of the investors' portfolio by y_j per unit, where j indexes the source of the risk.

Proposition 2 (Hedging and the sign of the dollar premium)**.** *From a zero-premium allocation with unhedged position n_0 and curvature K_0, an independent policy risk of variance V changes the dollar premium, in the absence of official transactions, by*

$$\ell_j(V) = -\frac{\Gamma \varrho r_j (y_j + n_0 r_j) V}{\Gamma + K_0 + \varrho r_j^2 V}. \tag{7}$$

The hedged position increases and the premium is negative when $r_j(y_j + n_0 r_j) > 0$; dollar demand rises when $r_j(y_j + n_0 r_j) < 0$, which occurs when the dollar's insurance payoff against other portfolio losses exceeds the risk of the existing dollar position.

The same preferences and dealer technology produce a stronger dollar when the risk is global and the dollar insures against it, and a weaker dollar when the risk originates in the United States and the dollar position adds to it. With many currencies, a premium that originates in U.S. policy risk moves the dollar against all of them by a common amount, unrelated to each currency's covariance with global risk, while the trade block sets the relative movements (equation (B.16) in Appendix B.3). A global risk produces the familiar ranking in which the dollar appreciates most against the currencies that fall in financial-market declines. Proposition 2 thus predicts the structure of the dollar's response rather than its size, and Section 5 takes that structure to 19 currencies. The tariff and portfolio components together form the dollar's pre-intervention innovation U.

3.3 The anchor rule and its choice

The authority responds to U with a sterilised purchase I of dollar claims against renminbi claims, priced in the same market. With price impacts $\beta \geq 0$ on the dollar against the basket and $\gamma > 0$ on effective renminbi depreciation (Appendix B.4),

$$B = U + \beta I, \qquad F = \chi + \gamma I, \qquad S = cB + F. \tag{8}$$

A rule $I = -kB$ implements

$$S = aB + \chi, \qquad F = (a - c)B + \chi, \qquad a = c - \gamma k. \tag{9}$$

The rule is linear in the dollar's movement because the basket pillar of the central parity is linear in it: a parity that passes the overnight basket movement in full gives a loading close to c, and a parity that offsets it gives a small loading. The rule acts on the common dollar component and leaves the direct trade response χ intact. With $t = k/(1 + \beta k)$ and $g = \gamma + c\beta$, the equilibrium is $I = -tU$, $B = (1 - \beta t)U$, $S = (c - gt)U + \chi$ and $F = -\gamma tU + \chi$.

The authority chooses t for the coming interval of news to minimise

$$\mathcal{J}(t) = \frac{1}{2}\mathbb{E}[F^2] + \frac{\zeta}{2}\mathbb{E}[I^2] + \frac{\omega}{2}\mathbb{E}[S^2] + \frac{\kappa_s}{2}\mathbb{E}[(S_+)^2], \qquad S_+ = \max\{S, 0\}. \tag{10}$$

The first term penalises effective exchange-rate variability, and the second captures the cost of intervention. The weight $\omega \geq 0$ captures the value of bilateral exchange-rate stability for dollar-based finance and risk sharing, and $\kappa_s \geq 0$ is a strategic cost of bilateral depreciation against the anchor issuer, which arises when the issuer responds to such depreciation with trade measures or negotiating demands. This cost can be larger when the anchor issuer is also the trade opponent. Appendix B.5 links ω to the dollar's insurance value and κ_s to the expected lasting tariff $m(D)$.

Proposition 3 (The opponent's currency as an anchor)**.** *For $\zeta > 0$ and $\mathbb{E}[U^2] > 0$ the loss is strictly convex, and its interior optimum satisfies*

$$(\gamma^2 + \zeta)t\mathbb{E}[U^2] - \gamma\mathbb{E}[U\chi] - \omega g\mathbb{E}[US] - \kappa_s g\mathbb{E}[US_+] = 0. \tag{11}$$

For a pure common-dollar innovation ($\chi = 0$), let $\pi_+ = \mathbb{E}[U^2\mathbf{1}\{U > 0\}]/\mathbb{E}[U^2]$, $\lambda = \omega + \kappa_s\pi_+$ and $\mathcal{C} = \gamma g/(\gamma^2 + \zeta)$. The optimal loading satisfies

$$\frac{c}{a^*} - 1 = \lambda\mathcal{C}. \tag{12}$$

A fall in ω loosens the anchor and a rise in κ_s tightens it whenever $\pi_+ > 0$. With unchanged financial coefficients and π_+, the anchor tightens despite a fall in the dollar-related stability benefit if

$$\pi_+\Delta\kappa_s > -\Delta\omega. \tag{13}$$

Economies with little strategic exposure to the anchor issuer respond mainly to a fall in ω and loosen their anchors, as in Hassan et al. (2026); the trade opponent also responds to κ_s. Equation (12) turns each year's loading into a composite of the policy weights. On days without tariff news, the regression of Δs on Δb recovers a when the own component on those days is uncorrelated with the dollar's movement, and $c/a - 1$ then measures $\lambda\mathcal{C}$. Responses to dollar appreciation and depreciation distinguish the two policy weights in the model. When the authority can respond differently to dollar appreciation and depreciation, the strategic term governs only the response to dollar strength: the loadings on the two sides satisfy $c/a_+ - 1 = (\omega + \kappa_s)\mathcal{C}$ and $c/a_- - 1 = \omega\mathcal{C}$, and their ratio identifies κ_s/ω free of the financial coefficients (Corollary B.1). Two

other motives make different predictions. A competitiveness motive values effective depreciation when tariffs rise and loosens the anchor on dollar appreciation, whereas a financial-stability motive resists depreciation during outflows and binds most when the dollar is strong. Lastly, a tighter rule moves U.S.-specific dollar risk into the anchoring country's effective rate: a U.S.-specific shock that lowers the dollar before intervention, $U_U < 0$, produces effective depreciation $F_U = (a - c)U_U/[1 + (\beta/\gamma)(c - a)]$, which rises as a falls even though official purchases dampen the dollar's depreciation (Proposition B.2).

3.4 Calibration and model-implied magnitudes

Table 1 reports the parameters and their quantitative implications. The assigned parameters are those of JS: the elasticities, home bias, the real interest rate, the Taylor coefficient, the slope of the Phillips curve and the 2017 trade shares, which together imply the offsets κ_m and κ_x. The measured constants come from published weights and trade values: the floating-basket weight $c = 0.680$, China's weight in the broad dollar index and the 2024 trade shares used for the second round. The remaining parameters and response coefficients are estimated as follows. The dollar loading a comes from days without tariff news. The credibility scale $D_0 = 47.5$ bp is the only structural parameter fitted to announcement-window responses: it minimises the squared residuals of the pooled two-day own-component equation when the coefficient on the mixture prediction of equation (6) is fixed at one and a reversed announcement lasts 14 days on average. The spot test does not reject scales between 20 and 60 bp, and the joint spot–forward test for 2018–19 does not reject scales between 10 and 80 bp. The coefficients of each round's two-day responses on the curve's prediction, 0.98 in 2018–19 and 0.74 in 2025–26, measure how each round's prices compare with the common curve. The hedging parameters and policy weights are assessed through model restrictions: the hedging block delivers a sign condition and a cross-currency structure, which Section 5 tests, and the policy block is identified through the composite $c/a - 1$.

Panel D reports the implied exchange-rate responses. At the fitted scale, credibility is 0.66 for a 25 bp announcement, 0.49 for 50 bp, 0.32 for 100 bp and 0.19 for 200 bp, so the expected lasting response rises only from 16 to 38 bp while the announcement rises eightfold. A reversed tariff with a mean duration of 14 days moves the renminbi within two days by 0.2 percent of its permanent response and lowers the one-year forward premium by 0.4 bp per 100 bp of permanent exposure. With a mean duration of 90 days the two figures are 4.8 percent and 13.5 bp, and with a mean duration of a year 25.5 percent and 45.4 bp. The forward market therefore separates a temporary tariff path of weeks from one of months, which is the restriction that Table 4 tests.

The fit uses only prices on announcement days, so the share of announced tariffs that was eventually implemented is an untargeted moment. In both rounds the share that markets priced is close to the share in force at the end of the last year of announcements: 0.24 against 0.26 at the end of 2019 for the U.S. increases on China first announced in 2018–19, and 0.20 against 0.20 at the end of 2025 for those announced from November 2024 to December 2025 (Table 3, Panel C). The curve itself implies 0.25 and 0.27; second-round prices ran below it, at 0.74 of its prediction, by the margin by which implementation fell short. Section 4 returns to these comparisons and to the later fall of the second-round share after a judicial ruling.

For the anchor, equation (12) maps the loading into the composite $c/a - 1 = \lambda\mathcal{C}$, which rose from 0.34 in 2023 to 2.44 in 2025 and 2.68 in 2026 (Table 6). In 2025 dollar appreciation carried only 34 percent of the dollar's variance on days without tariff news, and the five-day coefficient of the dollar on the VIX, a gauge of the dollar's insurance value, lay between −4.0 and 4.7 bp per VIX point, against 12 to 19 in 2023 (Appendix Table C6). A lower insurance value lowers ω, and a smaller π_+ gives the strategic term less weight, so with unchanged financial coefficients

Table 1: Calibration

Parameter or object	Value	Source or target
A. Assigned: Jeanne and Son (2024)		
Intertemporal substitution ϵ_i; Frisch ϵ_ℓ	1.00; 0.33	JS, Table 1
Import and export elasticities ϵ_m; ϵ_x	1.00; 3.00	JS, Table 1
Home bias ω_H: United States; China	0.85; 0.81	JS, p. 16
Real rate r; Taylor ϕ; Phillips slope α	0.03; 1.50; 1.03	JS, Table 1
China in U.S. imports; exports, 2017	0.219; 0.098	JS, Table 3
U.S. in China's imports; exports, 2017	0.084; 0.222	JS, Table 3
Implied offsets κ_m: United States; China	0.296; 0.286	$\kappa_x = 1.000$
B. Measured		
Floating-basket weight c (dollar bloc w)	0.680 (0.320)	CFETS 2017 (Table A2)
China in the dollar index ν: 2018; 2023; 2025; 2026	0.158; 0.112; 0.109; 0.109	Federal Reserve H.10
China in U.S. imports; exports, 2024	0.135; 0.069	Census Bureau
U.S. in China's imports; exports, 2024	0.055; 0.123	China Customs
C. Estimated and set in this paper		
Dollar loading a: 2018–19; 2025–26	0.575; 0.192	No-news days (Table 9)
Credibility scale D_0 (bp)	47.5	Two-day responses (Figure 2)
not rejected: spot; joint, 2018–19	20–60; 10–80	Figures 2 and C2
Mean life of a reversal $1/\rho_s$ (days)	14	Set; 90 in Table 4
Round slope on the curve: 2018–19; 2025–26	0.98 (0.26); 0.74 (0.23)	Equation (2)
D. Implied magnitudes		
Credibility $p(D)$ at $D = 25, 50, 100, 200$ bp	0.66; 0.49; 0.32; 0.19	Proposition 1
expected lasting response $Dp(D)$ (bp)	16; 24; 32; 38	
Two-day share ψ_x, lives of 14/30/90/365 days	0.002; 0.006; 0.048; 0.255	Equation (B.7)
one-year forward premium (bp)	−0.4; −1.7; −13.5; −45.4	Equation (B.8)
Anchor composite $c/a - 1$: 2023; 2025; 2026	0.34; 2.44; 2.68	Table 6
Effective response to a 100 bp U.S. shock: 2023 (bp)	17.3 to 14.9	β/γ from 0 to 0.921
2026 (bp)	49.5 to 34.0	β/γ from 0 to 0.923
E. Identified through restrictions		
Hedging block: ϱ, η, Γ, r_j, y_j, n_0	Sign condition	Proposition 2
Price impacts β, γ; weights ω, κ_s, ζ	Composite	$c/a - 1 = (\omega + \kappa_s \pi_+)\mathcal{C}$

Notes: Panel A uses the JS parameters and trade shares; source locations are in the data-source inventory. Panel C reports the pooled estimate of D_0 with the response coefficient fixed at one; "not rejected" denotes bootstrap $p \geq 0.05$. Panel D reports responses per 100 bp of permanent own-component exposure. The U.S. shock lowers the dollar by 100 bp before intervention; the range allows β/γ to vary from zero to the bound in equation (B.20). Estimation and interpretation are described in Section 3.4.

$\lambda = \omega + \kappa_s \pi_+$ could rise only through a larger κ_s. Section 6 splits the composite by the side on which the anchor binds, and Panel D also reports the transfer of dollar risk that Section 7 discusses.

The assigned parameters matter mainly through the trade elasticities: ϵ_x sets both the export offset and the substitution across import sources, and ϵ_m the import offset. Appendix Table C7 re-estimates the main objects under alternative values. The first-round results and the credibility scale change little: the first-round response coefficient lies between 0.18 and 0.26, D_0 between 43 and 53 bp, and the first-round priced share between 0.23 and 0.31 against an implemented share of 0.26 or 0.27, with the upper ends from $\epsilon_m = 2$. Shifting home bias by 0.05 moves D_0 by 0.1 bp and leaves the coefficients and shares unchanged. Second-round magnitudes depend more on ϵ_x and on the trade shares, which govern how strongly substitution across origins compresses the very large second-round rates: the second-round coefficient ranges from 0.09 to 0.49, and the priced share from 0.09 to 0.51 against implemented shares between 0.15 and 0.31.

4 How much of the offset did markets price?

4.1 One fifth in both rounds

Table 2 reports the calibration coefficients of equation (2). The renminbi's own component responded to tariff news by 0.20 of the permanent JS prediction in 2018–19 and by 0.18 in 2025–26. The difference between rounds is −0.03 (standard error 0.10, $p = 0.859$), and the pooled coefficient is 0.19 (0.05), with a minimum detectable effect of 0.14. The bootstrap rejects full pass-through in 2018–19 and in the pooled sample ($p < 0.001$); in 2025–26 alone, with 13 news clusters, its p-value is 0.127. The shares of the offset priced at announcement are statistically indistinguishable across rounds with different tariff levels, dollar movements and exchange-rate management. Offshore and onshore response coefficients on matched dates are not statistically distinguishable (paired p-values of 0.470 and 0.885), suggesting that the onshore trading band does not account for the result. The one-year CNH forward premium moved by −0.47 bp and 1.61 bp on signed bilateral news, and the three-month dollar OIS rate by −0.03 bp and −0.27 bp: neither the expected policy-rate differential nor the dollar's policy rate responded detectably, at any forward maturity (Appendix Table C3).

The dollar response differs across the two rounds. Its coefficient on the JS prediction was −0.54 in 2018–19 and 0.10 in 2025–26: the dollar moved against its trade prediction in the first round and with a small fraction of it in the second. Section 5 examines the dollar response in more detail.

The two-day window is the paper's unit of measurement. Over five days the first-round coefficient rises to 0.37 (0.10) while the second-round coefficient stays at 0.19 (0.12), and the own component already moved by 0.15 (0.09) of the announced exposure in the five days before first-round announcements, against 0.01 (0.06) in the second round (Appendix Table C2 and Figure C1). The estimates suggest more anticipation and slower exchange-rate adjustment in 2018–19; in 2018 the parity passed 0.63 of each day's market gap into the next day's central parity (Section 6). In 2025–26 the estimated response changed little beyond two days. The two-day coefficient is stable to the loading used, to first announcements only, to a basket without Asian currencies and to the exclusion of windows that span holiday closures (Appendix Table C2).

4.2 Size, origin and implementation

Table 3 relates the response to announcement size and the tariff-imposing country. Panel A separates the two origins of the own-component exposure: U.S. tariffs on China's exports, which depreciate the renminbi in JS, and Chinese tariffs on U.S. goods, which appreciate it. The coefficient

Table 2: Own-component and dollar responses to tariff news

	2018–19	2025–26	Pooled
A. Calibration slopes on the permanent JS prediction			
Own component, μ	0.20 (0.06)	0.18 (0.08)	0.19 (0.05)
bootstrap p, $\mu = 1$	<0.001	0.127	<0.001
minimum detectable effect	0.16	0.22	0.14
round 2 minus round 1	−0.03 (0.10), $p = 0.859$		–
Dollar against the basket	−0.54 (0.25)	0.10 (0.03)	0.07 (0.03)
B. Offshore and onshore renminbi on matched dates			
Own component, CNH	0.24 (0.08)	0.17 (0.09)	–
Own component, CNY	0.20 (0.06)	0.16 (0.07)	–
paired equality, p	0.470	0.885	–
C. Signed bilateral news (bp)			
One-year CNH forward premium	−0.47 (2.42)	1.61 (2.09)	–
Three-month dollar OIS rate	−0.03 (0.34)	−0.27 (0.18)	–

Notes: Panel A reports two-day coefficients from equation (2) for the own component and the dollar against the basket. The pooled estimate imposes a common coefficient. Minimum detectable effect: 2.8 times the CR1 standard error (5% significance, 80% power). Panel B compares CNH closing and CNY noon quotations on matched dates, conditional on period loadings. Panel C reports responses to signed bilateral news. The second-round sample in Panels B and C is 2025. CR1 standard errors are in parentheses; p-values use the bootstrap described in Section 2.

on U.S. measures is 0.20 and 0.14 in the two rounds and 0.18 pooled. Chinese measures move the renminbi against their own statutory effect, with a pooled coefficient of −0.62 (0.24), consistent with markets interpreting retaliation as evidence that U.S. tariffs would persist. Because Chinese measures are small by construction, a split of all announced exposures by size mixes size with origin (Appendix Table C1), so we test the effect of announcement size within U.S. measures.

Panel B tests whether credibility declines with announcement size. Within U.S. measures on China, the smaller half of announcements moved the renminbi by 0.75 of the permanent prediction in 2018–19 and the larger half by 0.14; the equality p-value is 0.009. Pooled across rounds, the coefficients are 0.53 and 0.13 ($p = 0.024$). The second round shows the same ordering with less precision (0.24 against 0.12).

Figure 2 plots the two-day responses against the permanent prediction with the long-run part of the common credibility curve, $D/(1 + D/D_0)$. Fixing the mean duration of the temporary tariff path at 14 days and the coefficient on the resulting prediction at one, the scale that best fits both rounds is $D_0 = 47.5$ bp of permanent own-component exposure (Section 3.4). Credibility is then one half for a 47.5 bp announcement, one third for a 100 bp announcement and one fifth for a 200 bp announcement, so quadrupling a threat from 50 to 200 bp raises the expected response by about three fifths, from 24 to 38 bp. The spot equation alone does not reject scales between 20 and 60 bp at the 5 percent level, which is the band shown.

Panel C of Table 3 compares the credibility estimates with subsequent implementation. The eight U.S. increases first announced in 2018–19 sum to 60.2 percentage points of average additional tariff, and the tariff in force at the end of 2019 was 15.1 points higher than at the start of 2018; in units of the JS prediction the implemented share is 0.26. For these announcements the curve implies a exposure-weighted credibility of 0.25, and first-round prices, which ran at 0.98 of the curve, priced a share of 0.24. The six increases first announced between November 2024 and December 2025 sum to 211.2 points, and the tariff in force at the end of 2025 was 18.4 points higher than before the first of them; in units of the JS prediction, which compresses the largest rates, the implemented share is 0.20. The curve implies 0.27 for them, while second-round prices ran at 0.74 of the curve and priced 0.20. The implemented shares play no part in the fit, so the

Table 3: Announced exposure, origin and implementation

	2018–19	2025–26	Pooled
A. Slope by origin of the dose			
U.S. measures on China	0.20 (0.06)	0.14 (0.06)	0.18 (0.05)
Chinese measures on the U.S.	−1.02 (0.77)	−0.66 (0.21)	−0.62 (0.24)
news days, U.S. / Chinese doses	18 / 11	13 / 8	31 / 19
B. U.S. measures on China, split at the median dose (88, 85 and 85 bp)			
Smaller half	0.75 (0.11)	0.24 (0.21)	0.53 (0.12)
Larger half	0.14 (0.05)	0.12 (0.05)	0.13 (0.04)
equality, p	0.009	0.649	0.024
news days, smaller / larger	9 / 9	7 / 6	16 / 15
C. Announced, implemented and priced U.S. tariffs on China			
Tariff in force on	31 Dec 2019	31 Dec 2025	23 Jul 2026
First-announced increases, pp (number)	60.2 (8)	211.2 (6)	214.9 (8)
Change in the tariff in force, pp	15.1	18.4	10.7
JS own response: announced increases (bp)	1,070	723	753
JS own response: implemented change (bp)	284	143	90
Implemented share, JS units	0.26	0.20	0.12
Credibility implied by the curve, $\sum Dp(D)/\sum D$	0.25	0.27	0.29
Priced share: round slope × curve credibility	0.24 (0.06)	0.20 (0.06)	0.21 (0.07)

Notes: Panels A and B estimate equation (2) for U.S. and Chinese measures separately. Panel B splits U.S. announcements at the median absolute exposure in each sample. Panel C includes only first announcements of U.S. tariff increases on China. Implemented changes are measured from 1 January 2018 and 24 November 2024. Announced and implemented exposures use the permanent JS prediction, in bp. Curve credibility is the exposure-weighted value of $p(D)$ at $D_0 = 47.5$ bp; the priced share multiplies it by the corresponding round's response coefficient. CR1 standard errors are in parentheses; equality tests use bootstrap p-values.

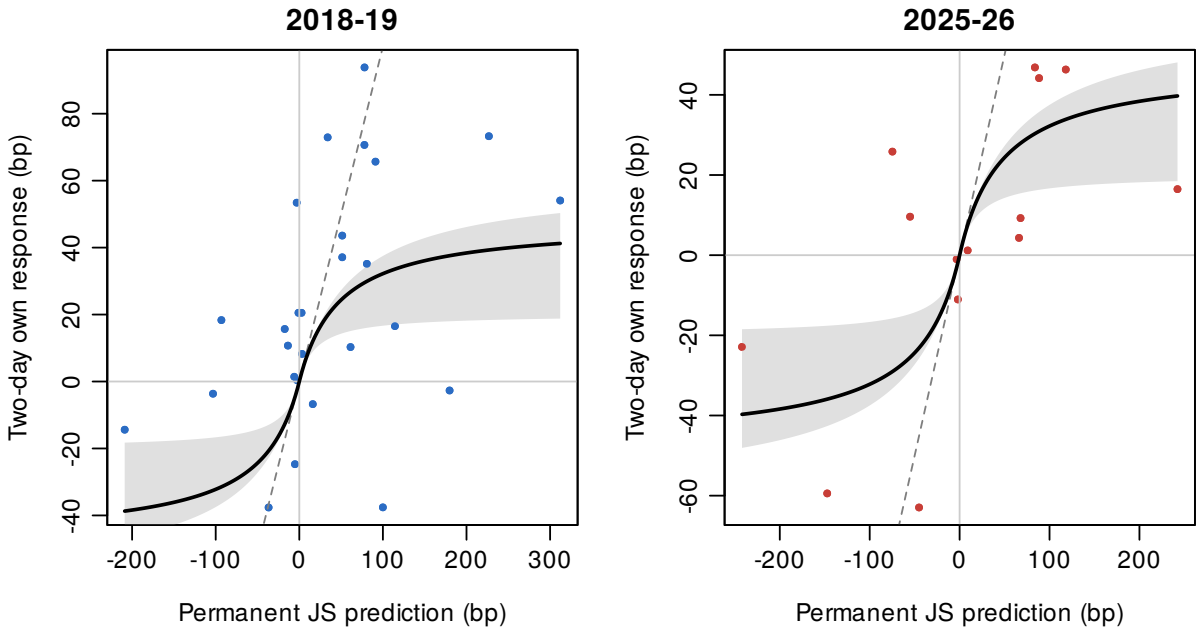


Figure 2: Announced exposure and the own-component response. *Notes:* Each point compares a news day's permanent JS prediction with the two-day own-component response (bp). The dashed line denotes full pass-through. The solid line is the long-run response implied by $D_0 = 47.5$ bp, estimated across both rounds with a mean temporary-tariff duration of 14 days. Shading covers the curves at tested scales not rejected by the pooled spot test at 5%. Estimation follows Section 3.4.

comparison is an untargeted test: the priced share was close to the implemented share, and in the second round they discounted announcements below the common curve by the margin by which implementation fell short. On 20 February 2026 the Supreme Court ruled that the International Emergency Economic Powers Act does not authorise tariffs (Supreme Court of the United States, 2026), and the emergency tariffs gave way to a temporary surcharge of 10 percent on all countries. With the two announcements of 2026 included, the implemented share on 23 July 2026 is 0.12, against a priced share of 0.21.

4.3 Spot and forward prices together

Table 4 stacks the own-component equation, with the coefficient on the mixture prediction restricted to one, and the one-year forward-premium equation, with the response restricted to its model value, and tests the two restrictions jointly with shared clusters. With a mean temporary-tariff duration of 14 days, a constant probability of permanence of 0.2 gives own-component coefficients of 1.01 and 0.82 and forward responses of −0.19 and −0.14 bp, with joint p-values of 0.992 and 0.863; the fitted credibility curve gives 0.995 and 0.510. A temporary tariff path with a mean duration of 90 days requires forward responses of −5 to −7 bp that are not observed, and the joint test gives $p = 0.064$ for the curve and 0.115 for the common probability in 2018–19, and 0.091 and 0.387 in 2025. Both restrictions set the financial premium's contribution to the policy-rate path to zero; the three-month dollar OIS rate moved by −0.03 and −0.27 bp on signed bilateral news (Table 2, Panel C), which provides no statistically detectable evidence of that contribution.

Table 4: Joint spot and forward restrictions

Path	Short branch	2018–19			2025		
	(days)	Spot slope	Forward (bp)	Joint p	Spot slope	Forward (bp)	Joint p
Common probability 0.2	14	1.01	−0.19	0.992	0.82	−0.14	0.863
Common probability 0.2	90	0.85	−6.56	0.115	0.69	−4.89	0.387
Credibility curve, $D_0 = 47.5$ bp	14	0.98	−0.19	0.995	0.66	−0.16	0.510
Credibility curve, $D_0 = 47.5$ bp	90	0.86	−6.56	0.064	0.60	−5.52	0.091

Notes: Predictions follow equation (6), using the stated mean temporary-tariff duration and either $p = 0.2$ or the estimated credibility curve. Spot coefficient: coefficient on the model prediction. Forward: predicted one-year CNH forward-premium response to signed bilateral news (bp). The joint bootstrap test assesses a spot coefficient of one and equality of observed and predicted forward responses. Equations use the same dates and clusters; the second-round sample is 2025. The financial premium's contribution to the forward curve is set to zero, as discussed in the text.

The first-round data are more informative about the credibility scale. Over a grid of scales with the temporary tariff path with a mean duration of 14 days, the joint test rejects $D_0 \leq 5$ bp and $D_0 \geq 120$ bp at the 5 percent level in 2018–19 and does not reject the range from 10 to 80 bp, which contains the fitted 47.5 bp (Appendix Figure C2). The second-round forward premia are consistent with every scale on the grid. The exchange-rate responses, variation in the size of U.S. announcements and implemented shares are consistent with the same interpretation: markets treated large tariff announcements as mostly temporary and priced the part that was likely to last.

5 The dollar component and the source of risk

Table 5 separates the three classes of tariff news. In 2018–19, U.S. measures on China moved the dollar by 8.3 bp against a JS prediction of 7.6 bp. Chinese retaliation moved the dollar up by 11.8 bp against a predicted fall of 24.3 bp ($p = 0.002$), a reversal of sign that a credibility discount cannot produce, because $p(D)$ only scales the prediction. In 2025–26, U.S. measures on

other partners raised the dollar by 12.2 bp against a prediction of 40.9 bp ($p = 0.008$). Chinese retaliation in 2025–26 lowered the dollar by 44.3 bp, close in size to the JS prediction of 52.3 bp; its largest days, in April 2025, fell inside the episode of broad dollar weakness described below. The two-year Treasury yield moved by a few basis points on any class of news, so the deviations from the JS prediction were accompanied by limited movements in this proxy for expected U.S. policy rates.

Table 5: The dollar's response to tariff news

News type	Dollar (bp)	JS (bp)	Equality p	2-year yield (bp)	VIX
A. Signed news, two-day window					
2018–19					
U.S. measures on China	8.3 (6.8)	7.6	0.923	−1.2 (1.2)	1.03 (0.42)
Chinese measures on the U.S.	11.8 (5.4)	−24.3	0.002	−1.7 (1.0)	0.24 (0.24)
U.S. measures on other partners	−4.6 (3.6)	5.6	0.055	0.6 (0.9)	−0.97 (0.18)
2025–26					
U.S. measures on China	14.5 (13.4)	25.2	0.639	−2.9 (1.1)	2.20 (0.88)
Chinese measures on the U.S.	−44.3 (15.0)	−52.3	0.608	2.9 (1.3)	−1.02 (0.73)
U.S. measures on other partners	12.2 (6.2)	40.9	0.008	−0.8 (1.5)	0.77 (0.84)

B. The dollar against 19 currencies on the largest announcement days					
Day (signed news)	Mean (bp)	Share down	Safe-haven slope	Tariff slope	Joint: safe-haven; tariff
20 Jul 2018 (+)	−67.9	0.95	1.1 [0.4]	–	–
06 May 2019 (+)	38.2	0.16	7.9 [3.4]	–	–
01 Jul 2019 (−)	29.7	0.21	−2.3 [−0.9]	–	–
02 Apr 2025 (+)	−123.7	0.84	4.8 [0.8]	4.2 [1.8]	12.1 [2.0]; 6.9 [2.5]
12 May 2025 (−)	44.1	0.26	−10.0 [−4.7]	4.3 [4.0]	−7.2 [−3.3]; 2.6 [2.9]
10 Oct 2025 (+)	28.3	0.47	10.9 [5.1]	−1.7 [−1.2]	12.0 [6.9]; 1.1 [1.1]
27 Oct 2025 (−)	−41.4	0.84	−0.4 [−0.2]	−0.8 [−0.9]	−1.7 [−1.2]; −1.2 [−1.3]
02 Apr 2025, first quote after release	−115.0	0.89	2.4 [0.4]	3.7 [1.9]	8.3 [1.8]; 5.5 [3.1]

Notes: Panel A reports two-day responses to the three news categories and the JS predictions estimated in the same regression. Dollar and Treasury-yield changes are in bp; VIX changes are in index points. CR1 standard errors are in parentheses; equality tests use bootstrap p-values. Panel B compares 19 currencies on the selected news days. "Share down" is the share against which the dollar depreciated. Safe-haven sensitivity is the 2017–2024 weekly exchange-rate coefficient on VIX changes. Tariff coefficients are in bp per percentage point of the announced reciprocal tariff. Joint estimates include both variables; HC1 t-statistics are in brackets. The final row uses noon quotations on 2 and 3 April 2025, bracketing the announcement at 16:06 Eastern Time on 2 April.

Panel B of Table 5 and Figure 3 show the cross-currency structure that Proposition 2 and equation (B.16) predict. Between the noon quotes of 2 and 3 April 2025, which bracket the release of the reciprocal tariffs, the dollar depreciated against 17 of 19 currencies, by 115 bp on average. It depreciated by similar amounts against the Mexican peso (272 bp), the yen (266 bp) and the Swiss franc (263 bp) (Appendix Table C4). Safe-haven status on its own does not explain the fall (coefficient 2.4, $t = 0.4$), and with the tariff rate held fixed its coefficient is 8.3 ($t = 1.8$). The dispersion around the common fall follows the announced tariffs: holding safe-haven status fixed, each percentage point of reciprocal tariff weakened a currency against the dollar by 5.5 bp ($t = 3.1$). This pattern is consistent with U.S.-specific risk weakening the dollar against most currencies and with the JS trade mechanism explaining relative currency movements.

The pattern differs on days on which the risk was bilateral. On 6 May 2019, the first trading day after the announced increase of the tariff on the USD 200 billion list to 25 percent, and on 10 October 2025, when the President threatened an additional 100 percent tariff on China, the dollar appreciated on average and appreciated most against the currencies that weaken in global financial-market declines (safe-haven coefficients of 7.9 and 10.9, with t-statistics of 3.4 and 5.1). The first round also had a day with the April 2025 pattern. On 20 July 2018, when the threat to extend tariffs to all Chinese imports came with the President's public criticism of the strong dollar and of the Federal Reserve's rate increases (Mohsin and Mayeda, 2018), the dollar depreciated

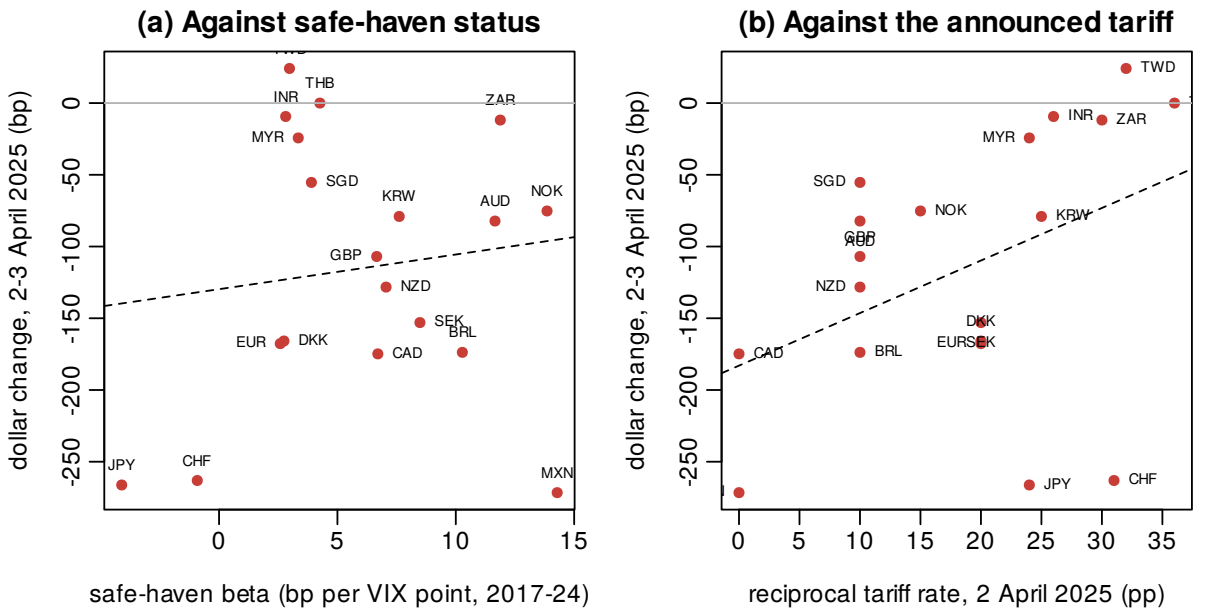


Figure 3: The dollar against 19 currencies after the reciprocal tariffs of 2 April 2025. *Notes:* Dollar changes against each currency between the H.10 noon quotations of 2 and 3 April 2025 (bp); negative values denote dollar depreciation. Panel (a) uses the currency's weekly response to VIX changes in 2017–2024. Panel (b) uses announced reciprocal tariff rates (Executive Order 14257, Annex I; 10% for unlisted partners, with Canada and Mexico exempt). Dashed lines are separate least-squares fits. Estimates and underlying data appear in Tables 5 and C4.

against 18 of 19 currencies, by 68 bp on average, again unrelated to safe-haven status (coefficient 1.1, $t = 0.4$). On 12 May 2025, the Geneva truce reversed the April pattern: the dollar appreciated by 44 bp, most against the safe havens (coefficient -10.0, $t = -4.7$). At lower frequency, the five-day coefficient of the dollar on the VIX, strongly positive in 2022–23 and again in 2026, lay between -4.0 and 4.7 bp per VIX point in 2025 across window alignments (Appendix Table C6 and Figure C3). The direction of the dollar response therefore varied with the source of risk. The renminbi's own-component response remained about one fifth of the permanent offset in both rounds.

6 The renminbi's anchor

6.1 A tighter dollar anchor, the largest shift among Asian currencies

Table 6 reports annual loading estimated on days without tariff newss for eight Asian currencies on the same calendar and basket. The renminbi's loading was 0.347 in 2017, rose to 0.629 in 2018, and fell to 0.470 in 2019. It stayed between 0.43 and 0.51 from 2020 to 2023 and then fell to 0.356 in 2024, 0.198 in 2025 and 0.185 in 2026. The anchor therefore loosened in the first year of the first round and tightened in the second; it tightened markedly as the second round approached. Figure 4 plots the path with a rolling 250-day estimate.

The timing is consistent with a role for the trade conflict. The Bai–Perron procedure (Bai and Perron, 2003; Zeileis et al., 2002), applied to the regression on days without tariff news with segments of at least 120 days and the number of breaks chosen by the Bayesian information criterion, selects a single break, with the new regime beginning on 13 November 2024 (95 percent interval 23 October 2024 to 4 March 2025) and loadings of 0.458 before and 0.189 after (Appendix

Table 6: Dollar loadings of Asian currencies, 2017–2026

Currency	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
CNY	0.347	0.629	0.470	0.444	0.429	0.466	0.508	0.356	0.198	0.185
	(0.040)	(0.055)	(0.064)	(0.056)	(0.042)	(0.051)	(0.039)	(0.044)	(0.039)	(0.030)
KRW	0.777	0.907	0.702	0.584	0.631	0.489	0.589	0.769	1.000	1.224
	(0.138)	(0.106)	(0.166)	(0.118)	(0.093)	(0.107)	(0.108)	(0.090)	(0.093)	(0.137)
TWD	0.631	0.632	0.485	0.301	0.349	0.363	0.452	0.534	0.483	0.254
	(0.078)	(0.065)	(0.126)	(0.049)	(0.048)	(0.054)	(0.041)	(0.060)	(0.210)	(0.056)
SGD	0.746	0.792	0.679	0.728	0.754	0.608	0.668	0.721	0.676	0.606
	(0.027)	(0.031)	(0.028)	(0.044)	(0.035)	(0.024)	(0.022)	(0.019)	(0.037)	(0.026)
MYR	0.141	0.298	0.229	0.476	0.230	0.269	0.365	0.748	0.388	0.459
	(0.034)	(0.047)	(0.130)	(0.057)	(0.053)	(0.066)	(0.051)	(0.156)	(0.077)	(0.095)
THB	0.419	0.793	0.634	0.573	0.555	0.805	1.011	1.079	0.750	0.955
	(0.035)	(0.082)	(0.091)	(0.063)	(0.069)	(0.057)	(0.070)	(0.110)	(0.092)	(0.119)
INR	0.367	0.678	0.415	0.470	0.560	0.312	0.165	0.080	0.181	0.165
	(0.067)	(0.128)	(0.146)	(0.087)	(0.086)	(0.048)	(0.039)	(0.019)	(0.058)	(0.082)
JPY	1.046	0.189	0.156	0.322	0.451	0.786	0.788	1.249	0.957	0.798
	(0.116)	(0.087)	(0.117)	(0.101)	(0.065)	(0.121)	(0.116)	(0.123)	(0.158)	(0.137)
CNY composite, $c/a - 1$	0.96	0.08	0.45	0.53	0.59	0.46	0.34	0.91	2.44	2.68
Dollar rises, π_+	0.35	0.57	0.47	0.51	0.64	0.50	0.45	0.62	0.34	0.42
Dollar, Δb (%)	−9.91	3.74	−0.30	−5.24	7.23	6.69	0.05	8.01	−7.24	−0.86

Notes: Annual regressions of daily bilateral exchange-rate changes on dollar changes against the fixed basket, using days without tariff news. Zero denotes a dollar peg; one denotes movement with the basket. Basket members are excluded from their own reference basket. HC1 standard errors are in parentheses; 2026 covers January–August. The final rows report $c/a - 1$ with $c = 0.680$, the share of squared dollar changes attributable to appreciation (π_+), and annual dollar changes in percent (2017 starts on 3 January).

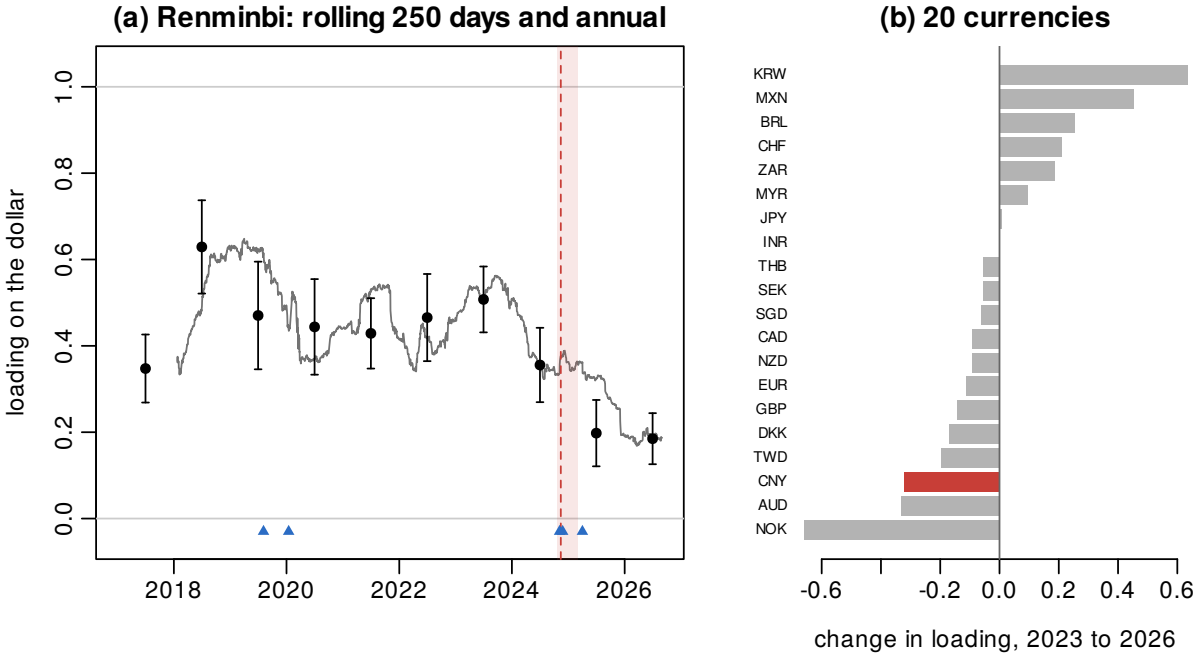


Figure 4: The renminbi's dollar loading and the cross-section of anchors. *Notes:* Panel (a): rolling estimates from days without tariff news within the preceding 250 common trading days (grey); annual estimates with 95% confidence intervals (points); estimated break date and 95% interval (dashed line and shading); institutional dates from Table C5 (triangles). Panel (b): changes in annual loadings from 2023 to 2026 for 20 currencies, excluding each currency from its reference basket where applicable.

Table C5). Over the 250 days without tariff news on either side, the loading fell from 0.374 to 0.181 around the first tariff threat of the second round on 25 November 2024 ($p < 0.001$), and from 0.601 to 0.381 around the manipulator designation of 5 August 2019 ($p = 0.004$); it did not change around the Phase One signature ($p = 0.861$). The first round thus produced a loosening in 2018 and a tightening once bilateral depreciation became a declared U.S. grievance, consistent with the change in κ_s that equation (B.21) in Appendix B.5 allows at that date.

The cross-currency evidence does not indicate a common regional pattern or a relationship with tariff exposure. The Korean won's loading rose from 0.589 in 2023 to 1.224 in 2026, the Malaysian ringgit's rose and the Indian rupee's did not change; the New Taiwan dollar's fell from 0.452 to 0.254, all of it in 2026. No other Asian currency has its best single break near November 2024, and the renminbi's fall of 0.323 is the largest among the Asian currencies of Table 6. Exposure to the reciprocal tariffs does not account for the tightening: across the 19 other currencies the change in the loading from 2023 to 2026 is uncorrelated with the tariff rates of April 2025 (correlation 0.08; Appendix Table C4). What sets China apart in the model is that bilateral depreciation against the anchor issuer had itself become an object of the dispute, through the designation of 2019 and the commitments of 2020, which can raise κ_s beyond the effect of tariff exposure common to trading partners.

6.2 The tightening through the central parity

Table 7 shows where the tightening was implemented. On days without news, the daily change in the central parity is regressed on the previous day's gap between the New York noon rate and the parity and on the previous day's move of the dollar against the basket. In 2017 and 2018 the parity passed 0.66 and 0.63 of the market gap into the next day's central parity, as the two-pillar rule describes. From 2023 the coefficient fell to 0.06, and in 2024–26 to between 0.01 and 0.02: the central parity became less responsive to the market. Its response to the dollar's movement against the basket fell from 0.31 in 2023 to 0.06 in 2025 and 0.05 in 2026, indicating reduced transmission of dollar movements to the central parity. Once the response to the market gap became small, from 2023, this coefficient is the parity's counterpart of the loading in the rule of Section 3.3: a parity that passed the basket movement in full would carry a coefficient near $c = 0.680$, and one that offset it would carry zero. Figure 5 plots both coefficients.

The market–central parity gap indicates whether the central parity resisted appreciation or depreciation. It averaged 52 bp in 2023, 109 bp in 2024 and 52 bp in 2025, a parity stronger than the market that leaned against depreciation. In 2026, with the renminbi appreciating against the dollar, it averaged −48.5 bp: the parity was set weaker than the market and leaned against appreciation. Panel B of Table 7 shows lower estimated loadings in both directions, with a smaller point estimate on dollar appreciation days. In 2025–26 the loading was 0.13 on days when the dollar appreciated and 0.24 on days when it depreciated ($p = 0.146$), both far below the 0.46 and 0.50 of 2022–23 ($p = 0.700$). Corollary B.1 reads the two loadings as the composites $(\omega + \kappa_s)\mathcal{C}$ and $\omega\mathcal{C}$ in the last column. Their ratio, less one, gives point estimates of κ_s/ω of 1.31 in 2025–26 and 0.32 in 2022–23: the estimated strategic weight relative to the stability weight was about four times as large, although the loading asymmetry is not statistically significant. From 2022 onwards the loading on dollar appreciation lies below that on dollar depreciation in every period of the panel, as a positive strategic weight implies. The two loadings were similar in 2018–19 (0.58 and 0.57), and in 2017, the year in which the counter-cyclical adjustment was introduced, the anchor was tighter on dollar depreciation.

The timing of the tightening also helps assess the strategic motive against the two alternatives of Section 3.3. A competitiveness motive would have loosened the anchor on dollar appreciation as

Table 7: The central parity and the symmetry of the anchor

A. Daily change in the central parity (bp)

Year	Previous gap	Previous dollar move	Mean gap (bp)	Days
2017	0.664 (0.053)	0.123 (0.028)	2.0	233
2018	0.635 (0.039)	0.119 (0.033)	0.6	193
2019	0.260 (0.042)	0.320 (0.044)	15.6	209
2020	0.543 (0.034)	0.178 (0.025)	−2.2	228
2021	0.725 (0.046)	0.111 (0.037)	−3.4	231
2022	0.364 (0.045)	0.273 (0.046)	17.6	230
2023	0.058 (0.019)	0.306 (0.042)	52.3	232
2024	0.017 (0.016)	0.131 (0.050)	108.9	227
2025	0.011 (0.004)	0.062 (0.012)	52.0	171
2026	0.016 (0.033)	0.050 (0.017)	−48.5	141

B. Renminbi loading when the dollar rises and when it falls

Period	Dollar rises	Dollar falls	Equality p	$c/a - 1$: rises; falls
2017	0.44 (0.08)	0.29 (0.07)	0.262	0.55; 1.36
2018–19	0.58 (0.06)	0.57 (0.08)	0.933	0.17; 0.19
2022–23	0.46 (0.05)	0.50 (0.06)	0.700	0.47; 0.36
2024	0.33 (0.07)	0.39 (0.08)	0.675	1.04; 0.76
2025–26	0.13 (0.04)	0.24 (0.05)	0.146	4.37; 1.89

Notes: Panel A relates daily central parity changes to the previous common day's market–central parity gap and dollar change against the basket (all in bp). Days with tariff news on that day or the preceding day are excluded. Positive mean gaps indicate a central parity stronger than the market. Panel B estimates loadings separately for dollar appreciation and depreciation days. The final column reports the policy-weight combinations in Corollary B.1. HC1 standard errors are in parentheses; equality tests use HC1 Wald p-values. The 2026 sample covers January–August.

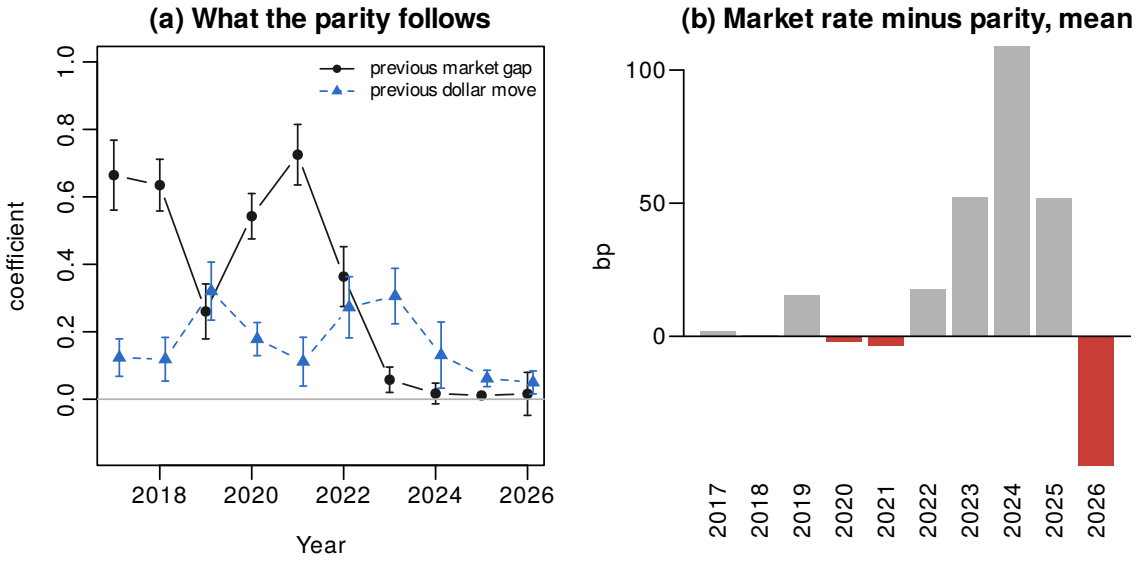


Figure 5: What the central parity follows. *Notes:* Panel (a): coefficients from Table 7, Panel A, with 95% confidence intervals. Panel (b): average H.10 noon rate minus central parity (bp); positive values indicate a central parity stronger than the market, and negative values indicate a weaker central parity.

tariffs rose, whereas the loading on those days fell from 0.46 to 0.13. A financial-stability motive binds most when the dollar is strong. The dollar appreciated against the basket by 7.23, 6.69 and 8.01 percent in 2021, 2022 and 2024, years in which the composite $c/a - 1$ was 0.59, 0.46 and 0.91; in 2025 it depreciated by 7.24 percent, and the composite reached 2.44, about two and a half times its previous peak of 0.96 (Table 6). That earlier peak came in 2017, when the dollar depreciated by 9.91 percent, but it rested on dollar depreciation, the side that the stability weight governs, whereas the composite of 2025–26 rests mainly on dollar appreciation. In 2023 the parity was already set 52 bp stronger than the market on average, yet it still passed 0.31 of the dollar's movement into the central parity: resistance to depreciation then worked through the level of the parity rather than through the loading. The anchor reached its tightest in 2025–26, in years of dollar weakness and after bilateral depreciation had become an object of the trade dispute, as the strategic rule predicts.

6.3 The prior loading governs transmission

The rule implies that the loading in force before an announcement governs how the announcement's dollar movement enters the bilateral rate. We estimate

$$S_t(h) = \alpha_r + \vartheta_1 B_t(h) + \vartheta_2 a_{t^-} B_t(h) + \text{signed-news controls} + u_t(h), \tag{14}$$

where a_{t^-} uses the days without tariff news among the preceding 250 common days; the rule predicts $\vartheta_1 = 0$ and $\vartheta_2 = 1$. Table 8 reports the estimates. On days without tariff news, $(\vartheta_1, \vartheta_2)$ is $(0.08, 0.80)$, close to the rule and consistent with attenuation from estimation error in the regressor. Over five-day news windows the estimates are -0.07 and 1.02 (joint $p = 0.634$), close to the rule. In the two-day window the coefficient on the prior loading is 3.23 (1.23) with $\vartheta_1 = -0.85$ (0.43); leaving out one news cluster at a time keeps it between 2.41 and 4.20, so news days pass more of the dollar's movement into the bilateral rate when the prior loading is high and less when it is low, before settling at the rule over the week. The joint test does not reject the rule at either window ($p = 0.494$ and 0.634), and at the mean prior loading of 0.398 the implied two-day loading is 0.435. The own-component coefficients are also statistically indistinguishable across the two rounds whose loadings differ by a factor of three, 0.20 and 0.18 with an equality p-value of 0.859 (Table 2), and it does not change when the window is measured with the prior loading in place of the period loading (Appendix Table C2), as implication (T4) of Appendix B.7 requires.

Table 8: Transmission predicted by the loading in force before the news

Sample	B	$a_{t^-}B$	Joint p	Implied / prior a	$a_{t^-}B$, one cluster out	Days / clusters
No news, daily	0.08 (0.06)	0.80 (0.14)	0.368	0.427 / 0.433	–	1919 / 131
All news, two days	−0.85 (0.43)	3.23 (1.23)	0.494	0.435 / 0.398	2.41 to 4.20	84 / 31
All news, five days	−0.07 (0.14)	1.02 (0.39)	0.634	0.331 / 0.398	0.88 to 1.36	84 / 31
Other-partner news, two days	0.09 (0.48)	0.33 (1.56)	0.619	0.195 / 0.315	−1.77 to 1.60	22 / 15

Notes: Equation (14) relates the bilateral change S to the dollar change B and its interaction with the loading estimated before the announcement, a_{t^-}. All regressions include period intercepts; news regressions also control for signed tariff news. The joint bootstrap test assesses coefficients of zero on B and one on $a_{t^-}B$. Implied loading is evaluated at the sample mean of a_{t^-}. "One cluster out" reports the range when each news cluster is omitted in turn. CR1 standard errors are in parentheses. The sample starts in 2018.

7 Where the adjustment went

Table 9 evaluates the identity (1) at the observed period changes, holding the round's loading fixed. This decomposition describes the observed adjustment; the stability of the own-component

coefficient across rounds with very different loadings (Table 2) connects it to the equilibrium of Section 3 for the announcement windows. In 2018–19 the dollar appreciated by 3.43 percent against the basket; at the high first-round loading it contributed −0.36 percentage points to the effective rate, and the own component's depreciation of 4.79 percent carried the effective depreciation of 4.43 percent. In 2025 the dollar depreciated by 7.24 percent. At the second-round loading of 0.192 that fall contributed 3.54 percentage points of effective renminbi depreciation, the own component contributed 2.90 points of appreciation, and the effective rate depreciated by 0.64 percent while the renminbi appreciated by 4.29 percent against the dollar. The bilateral and effective rates gave opposite readings of the same adjustment.

Table 9: Where the observed exchange-rate adjustment fell

Period	a	B	S	$(a-c)B$	x	F	x inside news windows	x outside news windows	Window days / all days
2018–19	0.575	3.43	6.77	−0.36	4.79	4.43	2.75	2.04	64 / 466
2025	0.192	−7.24	−4.29	3.54	−2.90	0.64	−0.16	−2.73	62 / 233
2026 (to 23 July)	0.192	2.22	−3.24	−1.08	−3.67	−4.75	−1.52	−2.15	15 / 130

Notes: Cumulative log changes in percent. Positive B denotes dollar appreciation; positive S and F denote bilateral and effective renminbi depreciation. The decomposition uses $F = (a-c)B + x$, $c = 0.680$ and each round's loading. News windows include the announcement day and the following common trading day. The final row reports trading days inside these windows and in the full period.

The own component's appreciation in 2025 happened almost entirely outside tariff-news windows: −2.73 of the −2.90 points accrued on the 171 days outside the two-day news windows, and only −0.16 on the 62 window days. This appreciation offset most of the effective depreciation that the dollar's depreciation would have produced. From January to 23 July 2026 the dollar appreciated by 2.22 percent and the own component appreciated by 3.67 percent, so the effective rate appreciated by 4.75 percent.

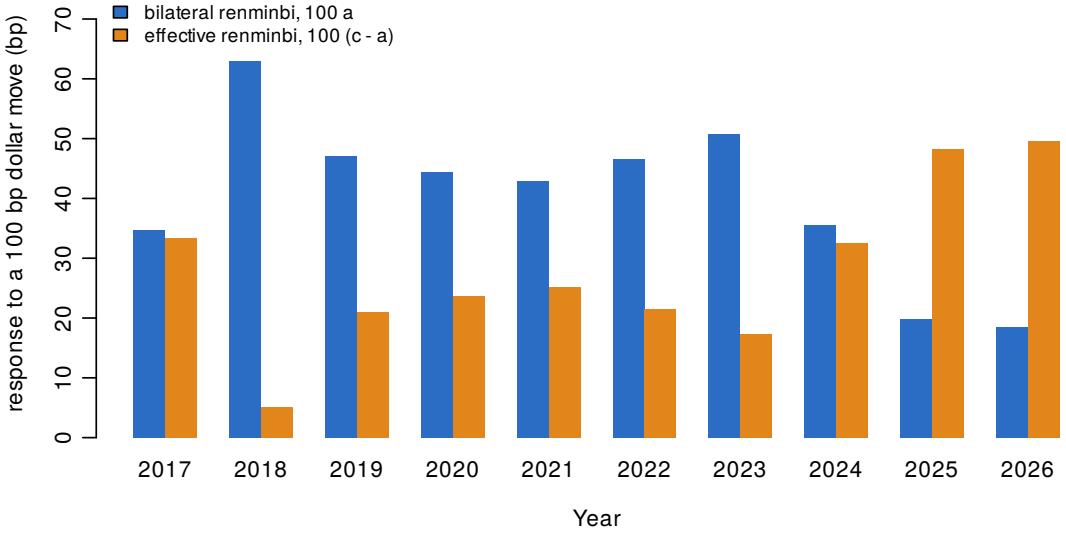


Figure 6: A tighter anchor changes where dollar movements are absorbed. *Notes:* Bilateral appreciation ($100a$) and effective depreciation ($100(c-a)$) implied by a 100 bp dollar depreciation against the basket, holding the own component fixed. Calculations use annual loadings from Table 6 and $c = 0.680$.

Figure 6 turns the loadings into magnitudes. At the 2018 loading, a 100 bp depreciation of the

dollar against the basket implies 62.9 bp of bilateral renminbi appreciation and only 5.1 bp of effective depreciation. At the 2023 loading the corresponding responses are 50.8 and 17.3 bp, and at the 2026 loading they are 18.5 and 49.5 bp: the effective-rate sensitivity to the dollar almost tripled in three years. Proposition B.2 adds the feedback of official purchases to the dollar. When official purchases move the dollar's effective value by no more than the renminbi's, that feedback is at most $\beta/\gamma = 0.92$ (Appendix B.4), so a U.S.-specific shock that would lower the dollar by 100 bp before official purchases raises effective renminbi depreciation by between 34.0 and 49.5 bp at the 2026 loading, against 14.9 to 17.3 bp at the 2023 loading (Table 1). Tighter management supports the dollar through official transactions but transfers more of the remaining movement into effective renminbi depreciation. The combination of U.S.-specific dollar risk and a strategically chosen anchor thus keeps the bilateral rate stable and moves the adjustment into the effective rate.

8 Conclusion and policy implications

This study examines exchange-rate adjustment when the tariff-imposing country also issues the anchor currency. Using 97 tariff announcements, the daily central parity and 20 currencies over 2017–2026, we find that the renminbi priced about one fifth of the permanent exchange-rate offset in both U.S.–China trade wars and that the response per unit of announced exposure fell with the size of U.S. announcements. The shares of announced increases that markets priced, 0.24 in the first round and 0.20 in the second, were close to the shares in force at the end of 2019 and of 2025, 0.26 and 0.20, which the calibration does not target. The dollar component followed the source of the risk: it rose on Chinese retaliation in 2018–19 and, on average, on the escalations of May 2019 and October 2025, and it fell broadly, as much against the peso as against the yen, following the broad U.S. tariff announcement of April 2025, consistent with U.S.-specific risk. China tightened its dollar anchor, from a loading of 0.508 in 2023 to 0.185 in 2026, through a central parity that became less responsive to market movements and resisted both depreciation and appreciation, and the loading in force before an announcement predicted its five-day transmission on average.

The identity of the anchor issuer connects these results. When that issuer is also the trade opponent, avoiding bilateral depreciation carries a strategic value that can rise while the anchor's value as insurance falls. The model derives the threshold at which the opponent tightens its anchor while other economies loosen theirs, and shows that the tighter rule transfers U.S.-specific dollar risk into the anchoring country's effective exchange rate. The timing of the anchor's breaks, the resistance of the parity on both sides with the lower loading on dollar appreciation, and the cross-section of anchors are consistent with an increase in the strategic weight: the anchor reached its tightest in 2025–26, while the dollar was weak.

These findings have implications for monetary authorities that anchor to the currency of a trade opponent. A tighter bilateral anchor during a period of dollar weakness raises the effective depreciation that the anchoring economy absorbs, so that bilateral stability comes with larger movements of the effective rate: at the 2026 loading, about half of a given dollar movement is transmitted to the renminbi's effective rate, holding the own component fixed. For tariff policy, announcements whose size exceeds what markets regard as sustainable move exchange rates by a fraction of their statutory size, and the fraction falls with the size of the threat. Assessments of tariff incidence should therefore account for the credibility of announced measures and the partner's exchange-rate policy alongside statutory tariff rates.

Intraday data would separate the announcement windows from the parity setting more finely, and positions and hedge ratios would quantify the private side of the dollar's response. A model of the announcement decision itself, in which governments choose the size of their threats knowing how markets discount them, would extend the analysis of trade wars between a country and the

issuer of its anchor currency.

Declarations

Data availability. The replication package contains every input file, one R entry point that writes each table and figure to its own file, the article source and a data-source inventory. The Bloomberg extract is supplied under the authors' licence for replication; all other inputs are public.

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the authors used Claude (Anthropic) to edit the language, to prepare and check the replication code, and to check the consistency between the text and the exhibits. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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A Data and events

This appendix lists every policy record, the basket weights, the statutory benchmark and summary statistics; Table 1 gives the JS parameters. The event identifier links each row of Table A1 to its source, coverage, announced rate, release-time record and repeated-announcement flag in `data/events.csv`. Retained records were checked against the official instrument (Federal Register notices, USTR, White House, MOFCOM and the State Council Tariff Commission) or, for statements, against the original post or transcript. A record is a repeat when it restates a measure already announced; Panel C of Table 3 and the sensitivity of Table C2 set the exposure associated with repeated announcements to zero. The data-source inventory of the replication package documents the publishers, the fixed releases and the checks of every input file.

Table A1: Announcement inventory and permanent predictions

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW1-034	2018-01-22	The President approves global safeguard tariffs on solar cells and modules (30%) and large residential washers (20% within quota, 50% above)	0 / 1	3.9	0.0
TW1-035	2018-03-01	The President announces tariffs of 25% on steel and 10% on aluminium imports from all countries (Section 232)	0 / 1	12.5	0.0
TW1-036	2018-03-08	The steel and aluminium proclamations are signed with Canada and Mexico exempted	0 / −1	−4.0	0.0
TW1-001	2018-03-22	USTR Section 301 findings released; the U.S. announces it will impose tariffs on Chinese imports	1 / 0	8.7	61.3
TW1-037	2018-03-22	The U.S. grants temporary exemptions from the steel and aluminium tariffs to the EU, Korea, Brazil, Argentina and Australia	0 / −1	−4.2	0.0
TW1-002	2018-04-02	China imposes retaliatory tariffs on U.S. aluminium scrap, pork, and other products	1 / 0	−4.1	−1.0
TW1-003	2018-04-03	USTR proposes 25% tariffs on about USD 50 billion of Chinese imports	1 / 0	7.3	51.4
TW1-004	2018-04-04	China announces proposed 25% tariffs on about USD 50 billion of U.S. goods	1 / 0	−72.1	−17.4
TW1-005	2018-04-05	The President directs USTR to consider tariffs on a further USD 100 billion of imports	1 / 0	14.1	100.0
TW1-006	2018-04-17	China imposes preliminary anti-dumping deposits on U.S. sorghum	1 / 0	−10.8	−2.6
TW1-007	2018-05-18	China terminates the sorghum measure during negotiations	−1 / 0	11.1	2.7
TW1-008	2018-05-29	The U.S. announces that the final USD 50 billion list will be published by 15 June	1 / 0	7.3	51.4
TW1-038	2018-05-31	The U.S. announces that the steel and aluminium exemptions for the EU, Canada and Mexico end on 1 June	0 / 1	5.9	0.0
TW1-009	2018-06-15	The U.S. and China each publish final USD 50 billion lists; first tranche effective 6 July	1 / 0	−64.9	33.9

Continued on the next page

Table A1 continued

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW1-010	2018-06-18	The President directs USTR to identify a further USD 200 billion of imports for 10% tariffs	1 / 0	11.4	80.8
TW1-011	2018-07-06	First tranches (USD 34 billion each) of U.S. and Chinese tariffs take effect	0 / 0	0.0	0.0
TW1-012	2018-07-10	USTR publishes the proposed USD 200 billion list	1 / 0	11.0	77.9
TW1-013	2018-07-20	The President threatens tariffs on all Chinese imports	1 / 0	32.0	226.8
TW1-014	2018-08-01	USTR considers raising the USD 200 billion tariff rate from 10% to 25%	1 / 0	16.1	114.4
TW1-015	2018-08-03	China announces proposed tariffs of 5-25% on USD 60 billion of U.S. goods	1 / 0	−56.5	−13.6
TW1-016	2018-08-07	USTR finalises the second USD 16 billion tranche of the USD 50 billion action	1 / 0	2.3	16.1
TW1-017	2018-08-08	China finalises 25% tariffs on USD 16 billion of U.S. goods (crude oil removed)	1 / 0	−21.4	−5.2
TW1-018	2018-08-23	Second USD 16 billion tranches take effect on both sides	0 / 0	0.0	0.0
TW1-019	2018-09-17	USTR finalises the USD 200 billion list at 10%, rising to 25% in 2019	1 / 0	25.4	179.8
TW1-020	2018-09-18	China finalises tariffs on USD 60 billion of U.S. goods at 5-10%	1 / 0	−25.1	−6.1
TW1-021	2018-09-24	U.S. USD 200 billion and Chinese USD 60 billion tariffs take effect	0 / 0	0.0	0.0
TW1-031	2018-12-01	At the G20 in Buenos Aires the U.S. agrees to keep the tariff on the USD 200 billion list at 10% instead of raising it to 25% on 1 January, during 90 days of negotiations	−1 / 0	0.0	0.0
TW1-032	2019-02-24	The President announces that the tariff increase scheduled for 1 March will be delayed, without a new date	−1 / 0	−16.7	−103.2
TW1-022	2019-05-05	The President announces that the tariff on the USD 200 billion list will rise from 10% to 25% on 10 May and that the remaining USD 325 billion of imports will soon be taxed at 25%	1 / 0	50.6	312.1
TW1-023	2019-05-13	China announces higher tariff rates on USD 60 billion of U.S. goods	1 / 0	−12.9	−3.1
TW1-039	2019-05-17	The U.S. agrees to lift the steel and aluminium tariffs on Canada and Mexico	0 / −1	−4.2	0.0
TW1-024	2019-06-01	China's higher tariffs on about USD 36 billion of goods take effect (weekend)	0 / 0	0.0	0.0
TW1-033	2019-06-29	At the G20 in Osaka the U.S. agrees not to impose tariffs on the remaining Chinese imports and to resume negotiations	−1 / 0	−33.8	−208.8
TW1-025	2019-08-01	10% tariffs announced on the remaining USD 300 billion of Chinese imports from 1 September	1 / 0	14.8	91.1

Continued on the next page

Table A1 continued

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW1-026	2019-08-13	USTR splits the USD 300 billion list into September and December tranches	−1 / 0	0.0	0.0
TW1-027	2019-08-23	China announces retaliatory tariffs of 5-10% on USD 75 billion of U.S. goods	1 / 0	−24.6	−5.9
TW1-040	2019-08-23	The U.S. responds by raising the rates on the USD 250 billion and USD 300 billion lists by 5 points	1 / 0	13.6	84.0
TW1-028	2019-09-11	China exempts 16 types of U.S. products from its retaliatory tariffs	−1 / 0	0.0	0.0
TW1-041	2019-09-11	The U.S. delays the increase from 25% to 30% on USD 250 billion of imports from 1 to 15 October	−1 / 0	0.0	0.0
TW1-029	2019-10-11	The U.S. suspends the October tariff increase ahead of the Phase One agreement	−1 / 0	−5.9	−36.7
TW1-042	2019-12-12	The President posts that the U.S. is getting very close to a big deal with China; the press reports the same day that the U.S. has offered to cancel the December tariffs and cut existing ones	−1 / 0	0.0	0.0
TW1-030	2019-12-13	Phase One agreement reached; the U.S. cancels the December tariffs	−1 / 0	−15.1	−93.2
TW0-001	2019-12-15	China suspends its 15 December tariffs on the second tranche of the USD 75 billion list and the restoration of tariffs on U.S. autos	−1 / 0	13.7	3.3
TW0-002	2020-01-15	Phase One agreement signed; USTR notice cuts the List 4A tariff from 15% to 7.5% from 14 February (implements 13 December announcement)	0 / 0	0.0	0.0
TW0-003	2020-02-06	China halves its tariffs on the first tranche of the USD 75 billion list (10% to 5%, 5% to 2.5%) from 14 February	−1 / 0	4.4	1.0
TW0-004	2020-02-14	U.S. and Chinese Phase One tariff reductions take effect (implementation)	0 / 0	0.0	0.0
TW0-005	2020-02-18	China opens a market-based exclusion process for U.S. goods subject to retaliatory tariffs (applications from 2 March)	−1 / 0	0.0	0.0
TW0-007	2021-10-04	USTR outlines a new China trade approach: tariffs kept, targeted process to reinstate 549 expired exclusions (notice 5 October)	−1 / 0	0.0	0.0
TW0-008	2022-03-23	USTR reinstates 352 of 549 exclusions from Section 301 tariffs, retroactive to 12 October 2021	−1 / 0	0.0	0.0
TW0-010	2024-05-14	The President directs higher Section 301 tariffs on about USD 18 billion of Chinese imports (EVs to 100%, semiconductors and solar cells to 50%, batteries to 25%)	1 / 0	1.7	7.6

Continued on the next page

Table A1 continued

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW0-011	2024-09-13	USTR finalises the four-year review increases from 27 September, with higher rates than proposed on some medical goods and proposed tariffs on tungsten, wafers and polysilicon	1 / 0	0.0	0.0
TW0-012	2024-09-27	First Section 301 increases from the four-year review take effect (implementation)	0 / 0	0.0	0.0
TW0-013	2024-11-25	The President-elect announces an additional 10% tariff on China (25% on Canada and Mexico) from his first day in office	1 / 1	245.1	85.0
TW0-014	2024-12-11	USTR finalises tariffs on tungsten (25%) and wafers and polysilicon (50%) from 1 January 2025, as proposed	0 / 0	0.0	0.0
TW2-048	2025-01-20	The tariffs promised for the first day in office (TW0-013) are not imposed: the America First Trade Policy memorandum orders reviews, and the President names 1 February as a possible start for 25% on Canada and Mexico (20 January) and 10% on China (21 January)	−1 / −1	0.0	0.0
TW2-001	2025-01-31	Executive orders announce 10% IEEPA tariffs on China (25% on Canada and Mexico) from 4 February	1 / 1	245.1	85.0
TW2-047	2025-02-03	The President pauses for one month the 25% tariffs on Mexico and for 30 days those on Canada ordered on 1 February; both now start at 12:01 a.m. on 4 March	0 / −1	0.0	0.0
TW2-002	2025-02-04	The 10% tariffs on China take effect; de minimis treatment ends (implementation)	0 / 0	0.0	0.0
TW2-003	2025-02-04	China announces retaliatory tariffs, export controls, and unreliable-entity listings	1 / 0	−5.5	−1.2
TW2-004	2025-02-07	The U.S. temporarily restores duty-free de minimis treatment for Chinese shipments	−1 / 0	0.0	0.0
TW2-005	2025-02-10	Proclamations impose 25% Section 232 tariffs on steel and aluminium from all sources	0 / 1	17.0	0.0
TW2-006	2025-02-27	The President announces that China will be charged an additional 10% tariff from 4 March (IEEPA tariff from 10% to 20%; executive order of 3 March)	1 / 0	15.5	67.8
TW2-007	2025-03-03	China announces retaliation: tariffs on U.S. farm exports, export controls, and other measures	1 / 0	−8.6	−1.9
TW2-049	2025-03-03	The President says there is no room left for Mexico or Canada: the paused 25% tariffs take effect on 4 March (end of the pause of TW2-047)	0 / 1	0.0	0.0

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Table A1 continued

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW2-050	2025-03-06	Executive orders exempt goods that qualify under the USMCA from the 25% tariffs on Canada and Mexico from 7 March (a one-month exemption for USMCA autos was announced on the afternoon of 5 March); non-USMCA potash falls to 10%	0 / −1	−103.5	0.0
TW2-008	2025-03-10	China's retaliatory tariffs take effect (implementation)	0 / 0	0.0	0.0
TW2-009	2025-03-12	Section 232 steel and aluminium tariffs take effect (implementation)	0 / 0	0.0	0.0
TW2-010	2025-03-26	Proclamation announces 25% Section 232 tariffs on automobiles and parts	0 / 1	65.3	0.0
TW2-011	2025-04-02	Executive order imposes a 10% baseline tariff on nearly all countries and country-specific reciprocal rates (34% for China)	1 / 1	339.1	118.1
TW2-012	2025-04-02	Executive order ends duty-free de minimis treatment for Chinese shipments from 2 May	1 / 0	0.0	0.0
TW2-013	2025-04-03	Section 232 tariffs on automobiles take effect (implementation)	0 / 0	0.0	0.0
TW2-014	2025-04-04	China announces 34% tariffs on all U.S. imports from 10 April and new export controls	1 / 0	−109.1	−24.4
TW2-015	2025-04-05	The 10% baseline tariff takes effect (weekend implementation)	0 / 0	0.0	0.0
TW2-016	2025-04-07	The President announces an additional 50% tariff on China from 9 April unless China withdraws its 34% retaliation (executive order of 8 April)	1 / 0	33.0	112.8
TW2-017	2025-04-09	Country-specific reciprocal tariffs take effect (implementation)	0 / 0	0.0	0.0
TW2-018	2025-04-09	Reciprocal tariffs paused for 90 days for other countries; the China rate is raised to 125%	1 / −1	−195.2	97.3
TW2-019	2025-04-09	China raises its retaliatory tariff on U.S. goods to 84%	1 / 0	−139.5	−31.2
TW2-020	2025-04-10	U.S. tariff changes of 9 April take effect (implementation)	0 / 0	0.0	0.0
TW2-021	2025-04-10	China's retaliatory tariffs of 4 April take effect (implementation)	0 / 0	0.0	0.0
TW2-022	2025-04-11	Smartphones and other electronics exempted from reciprocal tariffs	−1 / 0	−20.0	−68.2
TW2-023	2025-04-11	China raises its retaliatory tariff on U.S. goods to 125%	1 / 0	−29.7	−6.6
TW2-024	2025-04-12	China's tariff changes of 11 April take effect (weekend implementation)	0 / 0	0.0	0.0

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Table A1 continued

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW2-051	2025-04-13	The President says there was no tariff exception for electronics on 11 April, which move to a different tariff bucket; the Commerce Secretary says semiconductor tariffs will follow in a month or two (partial reversal of TW2-022)	1 / 0	0.0	0.0
TW2-052	2025-04-22	The President says the 145% tariff on China will come down substantially but will not be zero; the Treasury Secretary says current levels are not sustainable and expects a de-escalation	−1 / 0	0.0	0.0
TW2-053	2025-04-29	An executive order stops tariffs from stacking and the auto tariff proclamation is amended to offset part of the tariff for U.S.-assembled vehicles (relief from TW2-010)	0 / −1	0.0	0.0
TW2-025	2025-05-12	Geneva joint statement: the U.S. lowers the reciprocal tariff on China from 125% to 10% for 90 days	−1 / 0	−55.1	−185.3
TW2-026	2025-05-12	China reduces its retaliatory tariffs on the U.S. correspondingly	−1 / 0	170.8	38.1
TW2-028	2025-05-30	The President announces that the Section 232 tariffs on steel and aluminium will double from 25% to 50% (proclamation of 3 June)	0 / 1	28.2	0.0
TW2-029	2025-06-04	The 50% steel and aluminium tariffs take effect (implementation)	0 / 0	0.0	0.0
TW2-030	2025-06-11	The President announces a U.S.-China framework agreement after the London talks	−1 / 0	0.0	0.0
TW2-031	2025-07-07	The 90-day reciprocal-tariff pause is extended and new rates are set for other countries	0 / −1	0.0	0.0
TW2-044	2025-07-08	The President announces a 50% tariff on copper imports (Section 232; proclamation of 30 July, effective 1 August)	0 / 1	4.3	0.0
TW2-032	2025-07-10	Letter to Canada raises the tariff on Canadian goods from 25% to 35% from 1 August	0 / 1	2.8	0.0
TW2-033	2025-07-31	Executive order adjusts reciprocal tariff rates for nearly all countries	0 / 1	52.7	0.0
TW2-034	2025-08-11	Executive order extends the U.S.-China tariff truce for a further 90 days	−1 / 0	0.0	0.0
TW2-045	2025-08-19	The Commerce Department adds 407 product categories to the steel and aluminium derivative tariffs (50% on metal content)	0 / 1	7.5	0.0
TW2-035	2025-09-05	Executive order modifies the scope of the reciprocal tariffs (product exemptions; procedures for trade agreements)	0 / −1	−6.2	0.0

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Table A1 continued

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW2-046	2025-09-25	The President announces tariffs of 100% on branded drugs, 25% on heavy trucks, 50% on kitchen cabinets and 30% on upholstered furniture from 1 October	0 / 1	7.6	0.0
TW2-042	2025-10-10	The President announces an additional 100% tariff on imports from China from 1 November, in response to China's export controls on rare earths	1 / 0	79.6	242.3
TW2-054	2025-10-12	After the 100% threat the President posts that the U.S. wants to help China, not hurt it; the Vice President (12 October) and the Treasury Secretary (13 October) say the tariff does not have to happen (softening of TW2-042)	−1 / 0	0.0	0.0
TW2-055	2025-10-17	The President says the 100% tariff on China is not sustainable (softening of TW2-042)	−1 / 0	0.0	0.0
TW2-043	2025-10-26	After the Kuala Lumpur talks the Treasury Secretary says the U.S. and China have agreed a framework and the threatened 100% tariff is off the table	−1 / 0	−79.7	−242.0
TW2-036	2025-10-30	After the Busan meeting the President announces that the fentanyl-related tariff on China falls from 20% to 10%; China announces that it will adjust its countermeasures (fact sheet of 1 November)	−1 / 0	−11.5	−55.1
TW2-037	2025-11-04	Executive order implements the legal details of the agreement with China	0 / 0	0.0	0.0
TW2-038	2025-11-05	China's Tariff Commission announces reductions of retaliatory tariffs under the agreement	0 / 0	0.0	0.0
TW2-039	2025-11-10	USTR suspends port fees on Chinese ships and tariffs on Chinese cranes	−1 / 0	0.0	0.0
TW2-040	2025-11-14	Executive order removes IEEPA tariffs on some agricultural products from all sources	0 / −1	−8.7	0.0
TW3-013	2026-01-14	Proclamation imposes a 25% Section 232 tariff on certain advanced semiconductors from 15 January	0 / 1	1.9	0.0
TW3-002	2026-02-20	Supreme Court strikes down the IEEPA tariffs; executive order ends them and a proclamation imposes a 10% Section 122 surcharge on all countries from 24 February	−1 / −1	−68.6	−66.1
TW3-003	2026-02-21	The President announces the Section 122 surcharge will rise from 10% to 15% (never implemented)	1 / 1	40.2	21.0
TW3-004	2026-02-24	IEEPA duty collection ends and the 10% Section 122 surcharge takes effect (implementation)	0 / 0	0.0	0.0

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Table A1 continued

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW3-015	2026-02-25	The Trade Representative says the Section 122 surcharge will rise to 15% only for some countries (softening of TW3-003; the 15% was never implemented)	0 / −1	0.0	0.0
TW3-016	2026-03-04	The Treasury Secretary says the 15% Section 122 surcharge will likely start this week (it never took effect)	0 / 1	0.0	0.0
TW3-014	2026-04-02	Proclamation restructures the Section 232 tariffs on steel, aluminium and copper (50% on articles almost entirely of metal, 25% on other derivatives) from 6 April	0 / 1	28.9	0.0
TW3-008	2026-05-16	After the Beijing summit, China announces agreement in principle on reciprocal tariff cuts on goods of equal value and a Board of Trade	−1 / 0	0.0	0.0
TW3-009	2026-05-19	China's commerce ministry details the summit outcome: reciprocal tariff cuts on goods worth USD 30 billion or more each way, with MFN rates or lower expected; U.S. tariffs on China not to exceed the Kuala Lumpur level	−1 / 0	6.6	−3.3
TW3-010	2026-06-02	USTR proposes Section 301 forced-labour tariffs of 10% or 12.5% on 60 economies to replace the Section 122 surcharge, 12.5% for China	1 / 1	2.7	9.0
TW3-011	2026-07-23	USTR finalises the forced-labour tariffs from 24 July; 12.5% for China as proposed, lower treatment for the EU, Taiwan, Japan, Korea and Switzerland	0 / −1	0.0	0.0
TW3-012	2026-07-24	The Section 122 surcharge expires and the forced-labour tariffs take effect (implementation)	0 / 0	0.0	0.0
Excluded candidates					
TW0-006	2020-04-30	The President says new tariffs on China are an option in response to COVID-19	No specified tariff		
TW0-009	2022-05-03	USTR launches the statutory four-year review of the Section 301 tariffs on China	Investigation without tariff decision		
TW0-015	2024-12-23	USTR initiates a Section 301 investigation of Chinese legacy semiconductors	Investigation without tariff decision		
TW2-056	2025-04-25	China exempts eight types of U.S. semiconductors from its 125% tariff without an official announcement	No official announcement		
TW2-027	2025-05-28	U.S. Court of International Trade rules against the IEEPA tariffs (legal ruling)	Direction of tariff news unspecified		
TW2-057	2025-05-29	The Court of Appeals for the Federal Circuit stays the judgment of the Court of International Trade against the IEEPA tariffs	Direction of tariff news unspecified		

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Table A1 continued

ID	Release date	Announcement	Labels	B^{JS}	χ^{JS}
TW2-059	2025-07-12	Letters set tariffs of 30% on the European Union and Mexico from 1 August	Outside multi-partner event scope		
TW2-060	2025-07-27	The U.S. and the European Union agree a framework with a 15% tariff on most EU goods	Outside multi-partner event scope		
TW2-061	2025-07-31	The President extends the current tariff terms for Mexico by 90 days instead of the 30% set for 1 August	Outside multi-partner event scope		
TW2-058	2025-08-29	The Federal Circuit (en banc, by seven votes to four) holds the IEEPA tariffs unlawful but withholds its mandate pending an appeal	Direction of tariff news unspecified		
TW2-041	2025-12-23	USTR Section 301 notice on Chinese semiconductor policy; tariffs delayed	Direction of tariff news unspecified		
TW3-001	2026-01-12	The President announces a 25% tariff on any country doing business with Iran	Scope and China coverage unspecified		
TW3-018	2026-01-17	The President announces tariffs of 10% from 1 February, rising to 25% on 1 June, on Denmark, Norway, Sweden, France, Germany, the United Kingdom, the Netherlands and Finland	Outside multi-partner event scope		
TW3-019	2026-01-21	The President withdraws the tariffs on the eight European countries announced on 17 January	Outside multi-partner event scope		
TW3-005	2026-03-11	USTR initiates Section 301 investigations of excess capacity in 16 economies including China	Investigation without tariff decision		
TW3-006	2026-03-12	USTR initiates Section 301 investigations of forced-labour import bans in 60 economies including China	Investigation without tariff decision		
TW3-007	2026-05-07	U.S. Court of International Trade rules against the Section 122 surcharge; relief limited to the plaintiffs and stayed on appeal (legal ruling)	Direction of tariff news unspecified		
TW3-017	2026-05-12	The Federal Circuit stays the order of the Court of International Trade against the Section 122 surcharge	Direction of tariff news unspecified		
TW3-020	2026-07-20	Proclamations impose Section 338 tariffs of 50% on specified Canadian goods (about USD 20 billion) from 19 August	Outside multi-partner event scope		
TW3-021	2026-08-18	A proclamation temporarily suspends the Section 338 duties on Canadian goods, which take effect on 22 August instead of 19 August	Outside multi-partner event scope		
TW3-022	2026-08-22	The Section 338 tariffs on Canadian goods take effect	Outside multi-partner event scope		

Labels: bilateral / other-partner signs (+1 escalation, −1 de-escalation, 0 no news). B^{JS} and χ^{JS}: permanent JS predictions for the dollar against the basket and the renminbi's own component (bp). Exclusion reasons are

abbreviated; full reasons are in the event file. Each release is mapped to the first common trading day on or after its date.

Table A2: Official and estimation-basket weights

Currency	CFETS 2017 (%)	CFETS 2026 (%)	Fixed basket (%)
USD	22.40	18.31	–
EUR	16.34	17.86	25.13
JPY	11.53	8.12	17.73
KRW	10.77	8.46	16.56
AUD	4.40	5.41	6.77
HKD	4.28	4.06	–
MYR	3.75	5.15	5.77
SGD	3.21	3.00	4.94
GBP	3.16	2.74	4.86
THB	2.91	3.64	4.47
RUB	2.63	5.04	–
CAD	2.15	2.43	3.31
SAR	1.99	2.85	–
AED	1.87	2.61	–
ZAR	1.78	1.40	2.74
CHF	1.71	1.47	2.63
MXN	1.69	2.78	2.60
TRY	0.83	1.15	–
PLN	0.66	1.19	–
SEK	0.52	0.51	0.80
NZD	0.44	0.54	0.68
DKK	0.40	0.45	0.62
HUF	0.31	0.46	–
NOK	0.27	0.28	0.42
MOP	–	0.11	–
Dollar-bloc share w and $c = 1 - w$		0.320 and 0.680	

Notes: Official CFETS basket weights for 2017 and 2026 (percent). The estimation basket fixes the 2017 weights of the 16 floating currencies available in H.10 and rescales them to sum to one. The dollar bloc (USD, HKD, SAR and AED) has weight $w = 0.320$ in the quoted basket.

Table A3: Statutory benchmark and observed changes

	2018–19	2025	2026 (to 23 July)
A. Change in average additional tariffs (percentage points)			
U.S. on China	15.1	18.4	−7.7
U.S. on other countries	0.0	9.0	−1.2
China on the U.S.	14.2	11.1	0.0
B. Change in effective tariffs (percentage points)			
U.S. imports	2.80	9.92	−1.84
U.S. exports	1.16	0.48	0.00
China's imports	0.99	0.38	0.00
China's exports	2.84	1.43	−0.53
C. Exchange rates: JS prediction and observed change (percent)			
Dollar against the basket, Δb: JS	−0.78	2.40	−0.50
observed	3.43	−7.24	2.22
Effective renminbi, ΔE: JS	2.55	1.32	−0.53
observed	4.43	0.64	−4.75
Renminbi per dollar, Δs: JS (free float)	2.02	2.95	−0.87
observed	6.77	−4.29	−3.24
China's weight in the dollar index, ν	0.158	0.109	0.109
Import bases (USD bn): U.S. from China; China from the U.S.	525.7; 154.8	440.3; 143.3	440.3; 143.3
Offsets κ_m, κ_x: U.S.; China		0.296, 1.000; 0.286, 1.000	

Notes: Panel A: changes in import-weighted additional tariffs. Panel B: effective tariffs allowing substitution across import sources ($\epsilon_x = 3$). Panel C: permanent JS predictions and observed log changes (percent); the bilateral prediction assumes a floating renminbi ($a = c$). First-round trade shares and import bases follow JS; the second round uses 2024 trade values.

Table A4: Summary statistics

	2018–19	2020–24	2025–26
Common trading days	466	1,162	390
Tariff-news days	36	7	42
with U.S. measures on China	22	5	22
with Chinese measures on the U.S.	12	2	10
with U.S. measures on other partners	6	1	23
News clusters	13	6	13
Mean absolute daily change (bp): news days / no-news days			
s, news days	27.6	10.0	16.1
s, no-news days	20.2	19.7	10.8
b, news days	21.5	22.6	40.1
b, no-news days	20.1	29.2	28.5
x, news days	23.7	5.0	13.4
x, no-news days	16.8	16.3	9.3
Mean $\lvert\chi^{JS}\rvert$ on dose days (bp)	73.0	31.2	88.9
Mean $\lvert B^{JS}\rvert$ on dose days (bp)	18.8	83.7	63.8
Days with CNH forward quotes	466	247	233
Full sample, 3 January 2017 to 31 August 2026: 2,253 common trading days			
Policy records: 140 candidates, 119 retained, 21 excluded			
retained: 97 with tariff news, 22 implementation steps			

Notes: The sample uses common U.S.–China trading days. News days have non-zero signed labels; exposure days ("dose days") also have non-zero permanent predictions. News clusters group days within five common trading days of tariff news. Daily changes are in bp; x uses period loadings. CNH quotations cover 2017–19 and 2024–25. The final rows give full-sample event counts.

B The model: equilibrium and proofs

This appendix sets out the parts of the model that Section 3 summarises. Appendix B.1 gives the JS real block and its response operators, B.2 credibility and the identification of spot and forward prices, B.3 private hedging, B.4 official intervention and index aggregation, B.5 the anchor choice, B.6 the transfer of dollar risk, and B.7 the JS special case with the testable implications.

B.1 The JS block and its response operators

We retain the JS preferences, producer-currency pricing, quadratic price adjustment and the Taylor rule for home-goods inflation. The linearised block for each economy is

$$c_t^H = \bar{c} - \epsilon_i(\omega_H z_t + \omega_F \tau_t) - \epsilon_i \int_0^t \ell_u \, du, \tag{B.1}$$

$$y_t = \omega_H c_t^H - \omega_F[\omega_H \epsilon_m (z_t - \tau_t) + \epsilon_x (z_t + \tau_t^*)], \tag{B.2}$$

$$\dot{\widetilde{\pi}}_t = r\widetilde{\pi}_t - \alpha[y_t/\epsilon_\ell + c_t^H/\epsilon_i - \omega_F(z_t - \tau_t)], \tag{B.3}$$

$$\dot{z}_t = -(\phi - 1)\widetilde{\pi}_t - \ell_t, \qquad i_t = r + \widehat{\pi} + \phi\widetilde{\pi}_t. \tag{B.4}$$

Here z rises with real appreciation, τ is the effective home import tariff, τ^* the effective tariff on home exports and $\omega_F = 1 - \omega_H$. A positive ℓ is a premium on home-currency claims, $i_t + \ell_t = i_t^* + \dot{e}_t$, with e the home-currency price of foreign currency. With $\ell = 0$, an unanticipated permanent tariff gives $de = -\kappa_m d\tau + \kappa_x d\tau^*$ and no change in the policy rate. We write $\mathcal{L}_x[n]$ and $\mathcal{L}_H[n]$ for the own-currency and H-maturity forward-premium responses to news about the effective tariff path,

$$n(u) = \mathbb{E}[\tau_{t+u} \mid \mathcal{I}_t] - \mathbb{E}[\tau_{t+u} \mid \mathcal{I}_{t^-}]. \tag{B.5}$$

The household budget and goods-market equations are evaluated to first order around zero net foreign assets. Legacy gross dollar claims are held by the foreign portfolio sector. Same-currency counterclaims offset their initial contribution to the consolidated JS household sector's external wealth. The foreign investor and dealer budgets include their asset payoffs and forward settlements; a forward payment is a receipt for its counterparty. Intervention starts from zero, is financed with domestic bonds and is sterilised. Its realised profits are rebated. Products of a newly acquired position and a return innovation are second order and do not enter the first-order goods resource constraint. The portfolio problem retains its quadratic risk terms.

Define the JS constants

$$K_R = 1 + \frac{\epsilon_i \omega_H^2 + \omega_F(\omega_H \epsilon_m + \epsilon_x)}{\epsilon_\ell}, \qquad M_R = \frac{\omega_H}{\epsilon_\ell} + \frac{1}{\epsilon_i},$$

$$G_R = \omega_H \epsilon_m + \epsilon_x - 1 - \epsilon_i \omega_H, \qquad H_R = \epsilon_i \omega_F + \epsilon_m \omega_H,$$

$$\Theta_m = \frac{\omega_F \omega_H (\epsilon_m - \epsilon_i)}{\epsilon_\ell}, \qquad \Theta_x = \frac{\omega_F \epsilon_x}{\epsilon_\ell}.$$

For zero financial premia, the net-export condition is

$$nx_t/\omega_F = -\bar{c} - G_R z_t + H_R \tau_t - \epsilon_x \tau_t^*.$$

The present-value budget constraint $\int_0^\infty e^{-rt} nx_t dt = 0$ determines

$$\bar{c} = r \int_0^\infty e^{-rt}[-G_R z_t + H_R \tau_t - \epsilon_x \tau_t^*] dt.$$

For constant tariffs this gives

$$\kappa_m = \frac{M_R H_R + \Theta_m}{K_R + G_R M_R}, \qquad \kappa_x = \frac{M_R \epsilon_x + \Theta_x}{K_R + G_R M_R}.$$

For an expected tariff path proportional to $e^{-\rho t}$, put

$$\mathcal{D}(\rho) = \rho(r+\rho) + \alpha K_R(\phi - 1), \quad \delta_R = \frac{\alpha K_R(\phi-1)}{\mathcal{D}(\rho)}, \quad \eta_R = \frac{r}{r+\rho}.$$

The exchange-rate jump is $de_0 = -\kappa_m \psi_m \tau_0 + \kappa_x \psi_x \tau_0^*$, where

$$\kappa_m \psi_m = \delta_R \frac{\Theta_m}{K_R}\left(1 - \eta_R \frac{G_R M_R}{K_R + G_R M_R}\right) + \eta_R \frac{M_R H_R}{K_R + G_R M_R}, \tag{B.6}$$

$$\kappa_x \psi_x = \delta_R \frac{\Theta_x}{K_R}\left(1 - \eta_R \frac{G_R M_R}{K_R + G_R M_R}\right) + \eta_R \frac{M_R \epsilon_x}{K_R + G_R M_R}. \tag{B.7}$$

The bounded particular solution is $z_t = M_R \bar{c}/K_R + z_e e^{-\rho t}$, with $z_e = \delta_R(\Theta_m \tau_0 - \Theta_x \tau_0^*)/K_R$. Substitution in the budget condition proves these expressions. The two homogeneous roots have positive real parts for $\alpha > 0$ and $\phi > 1$, so boundedness removes the homogeneous terms.

The nominal change between impact and horizon h years is

$$de_h - de_0 = z_e(1 - e^{-\rho h})\frac{\phi}{\phi - 1}.$$

The code evaluates this correction at $h = 2/252$ for a two-common-day window. A foreign tariff on home exports changes the average policy rate over the next H years by

$$\Delta \bar{i}_H = -\frac{\phi \rho \alpha \Theta_x \tau_0^*}{\mathcal{D}(\rho)} \frac{1 - e^{-\rho H}}{\rho H}. \tag{B.8}$$

Under the maintained covered interest parity condition, the log forward premium is $H(\bar{i}_{CN,H} - \bar{i}_{US,H})$. Equation (B.8) applied to the two countries gives the trade-block operator $\mathcal{L}_H$. At the calibration $\epsilon_m = \epsilon_i = 1$, $\Theta_m = 0$, so the import tariff does not directly move the policy rate. For $\rho = 0$ both exchange-rate shares are one and the rate response is zero.

With a premium, let $L_t = \int_0^t \ell_u du$. Net exports become

$$nx_t/\omega_F = -\bar{c} + \epsilon_i L_t - G_R z_t + H_R \tau_t - \epsilon_x \tau_t^*.$$

Thus the external budget includes the premium-induced consumption term. For $\ell_t = \ell_0 e^{-\rho_\ell t}$, $\rho_\ell > 0$, the zero-tariff particular solution is $z_t = A_z + z_e e^{-\rho_\ell t}$, with

$$z_e = \left[r + \rho_\ell + \frac{(\phi-1)\alpha M_R \epsilon_i}{\rho_\ell}\right]\frac{\ell_0}{\mathcal{D}(\rho_\ell)}, \tag{B.9}$$

$$A_z = -\frac{M_R \eta_R}{K_R + G_R M_R}\left(\frac{\epsilon_i \ell_0}{\rho_\ell} + G_R z_e\right), \qquad \eta_R = \frac{r}{r+\rho_\ell}. \tag{B.10}$$

The impact multiplier is $\Lambda = (A_z + z_e)/\ell_0$. It is positive when $G_R \geq 0$ and $\Theta_m + \Theta_x > 0$, as at the calibration. To verify the sign, multiply Λ by the positive quantity $\rho_\ell \mathcal{D}(\rho_\ell)(K_R + G_R M_R)$. The numerator is

$$\begin{aligned}(r+\rho_\ell)\rho_\ell\left[(\Theta_m + \Theta_x) + (1-\eta_R)M_R(\epsilon_i + G_R)\right] \\ + (\phi-1)\alpha M_R \epsilon_i (1-\eta_R)(K_R + G_R M_R) > 0.\end{aligned}$$

The policy-rate contribution is

$$\Delta i_t = -\frac{\phi\alpha(\Theta_m + \Theta_x)\ell_0}{\mathcal{D}(\rho_\ell)} e^{-\rho_\ell t}. \tag{B.11}$$

Its H-maturity average is the impact value multiplied by $(1 - e^{-\rho_\ell H})/(\rho_\ell H)$. Sterilisation concerns the financing of intervention; equation (B.11) separately describes the endogenous Taylor-rule response. These rate terms enter the complete forward operator alongside equation (B.8); the forward tests of Section 4 set them to zero, because the estimated response of the three-month dollar OIS rate to tariff news was not statistically significant.

B.2 Credibility and the identification of spot and forward prices

Proof of Proposition 1. Continuation is optimal if $v \geq \delta D$. The survival function gives $p(D) = (1+\delta D)^{-1}$. Conditional on reversal, the probability that the initial change survives to u is $e^{-\rho_s u}$. Total expectation gives equation (4). The long-run increment is εDZ, where Z indicates continuation, giving $m(D)$ and $V(D)$. Direct differentiation yields

$$m'(D) = \frac{1}{(1+\delta D)^2}, \quad m''(D) = \frac{-2\delta}{(1+\delta D)^3}, \quad V'(D) = \frac{\delta D^2(3+\delta D)}{(1+\delta D)^3}.$$

Proposition B.1 (Spot and forward prices)**.** *Consider the trade block at a given financial-premium path and a fully unanticipated change. Let $G = X_\infty - X_s$ and let a prime denote a derivative with respect to ρ_s. At an interior $p \in (0,1)$, the spot response and one forward premium locally identify (p, ρ_s) whenever*

$$GP'_{H,s} + P_{H,s}X'_s \neq 0. \tag{B.12}$$

With $p = p(D)$, different announced exposures and maturities impose further restrictions on (δ, ρ_s). As the duration of the temporary tariff path approaches zero, the forward response vanishes and prices retain the product of the probability of permanence and the news content of the announced exposure.

An anticipated change enters through the revision of its expected path in equation (B.5), so the limit in Proposition B.1 is the relevant one when the temporary tariff path resolves within weeks. Three pieces of evidence then separate credibility from anticipation. Within exchange-rate responses, curvature with respect to announced exposure helps distinguish them under the maintained model, provided that it is estimated for a single tariff-imposing country, because U.S. and Chinese measures differ in predicted size by construction. The implemented share of announced tariffs provides a separate measure of realised persistence, and the own-component change in the days before each announcement measures anticipation. Section 4 reports all three.

Proof of Proposition B.1. The Jacobian of (X, P_H) with respect to (p, ρ_s) is

$$J = \begin{pmatrix} X_\infty - X_s & (1-p)X'_s \\ -P_{H,s} & (1-p)P'_{H,s} \end{pmatrix}.$$

Its determinant is $(1-p)[(X_\infty - X_s)P'_{H,s} + P_{H,s}X'_s]$. The inverse-function theorem gives local identification at a nonzero determinant. With several announced exposures, $\partial p/\partial\delta = -D/(1+\delta D)^2$ varies across observations. Distinct maturities supply distinct $P_{H,s}$ and derivatives. As $\rho_s \to \infty$, $X_s, P_{H,s} \to 0$, and the price system becomes $X = pX_\infty$, $P_H = 0$. An unknown multiplicative news share then enters only through its product with p. The grid tests of Section 4 retain every parameter point that satisfies the joint restriction.

B.3 Private hedging and the source of dollar risk

Foreign investors hold a predetermined dollar position A and choose the unhedged amount n; their hedge is $H = A - n$. Let R be the dollar excess return over the decision interval, before its expected premium $-\ell$, and Y the payoff on the rest of the investor's portfolio. The investor maximises

$$-\ell n - \frac{\varrho}{2}\operatorname{Var}(Y + nR) - \frac{\eta}{2}(n - A)^2, \qquad \varrho > 0,\ \eta > 0, \tag{B.13}$$

where the last term is a convex hedging cost, so that a hedge can change without a sale of the underlying asset. The unhedged position is

$$n = \frac{\eta A - \varrho \operatorname{Cov}(R, Y) - \ell}{K}, \qquad K = \eta + \varrho \operatorname{Var}(R). \tag{B.14}$$

A dealer chooses its dollar position q to maximise $-\ell q - \Gamma q^2/2$, with $\Gamma > 0$ the marginal cost of bearing currency risk (Gabaix and Maggiori, 2015). With official net dollar purchases I and net supply $\bar{N}$, market clearing $n + q + I = \bar{N}$ gives

$$\ell = \frac{\Gamma K}{\Gamma + K}(d + I), \qquad d = \frac{\eta A - \varrho \operatorname{Cov}(R, Y)}{K} - \bar{N}. \tag{B.15}$$

Private rebalancing and official transactions are priced in the same market. Let an additional policy innovation be $\epsilon = D(Z - p)$, where Z indicates continuation, so that its variance is $V(D)$, and write the payoff changes as $R = r_j \epsilon$ and $Y = y_j \epsilon$ for a source j of policy risk.

Terminal payoffs and the portfolio interval. At the resolution date, continuation and reversal specify two tariff allocations. Equations (B.1)–(B.4), with the financial innovation resolved, determine the two real allocations and currency values. For proportional packages, their differences give the local currency payoff coefficient r_j. The investor's predetermined claims on these allocations give y_j: for claims with quantities A_m and state-contingent non-currency payoff differences $d_{mj}D$, $y_j = \sum_m A_m d_{mj}$. Currency payoffs on dollar holdings enter through nR and are excluded from Y to avoid counting them twice. Negative and positive covariance with R are both possible across different portfolios and tariff packages. The policy authority chooses its rule for the current rebalancing interval, taking these terminal payoff schedules as given, and the calculation covers that one interval.

Proof of Proposition 2. The first-order condition of equation (B.13) is

$$-\ell - \varrho[n \operatorname{Var}(R) + \operatorname{Cov}(R, Y)] - \eta(n - A) = 0.$$

Strict concavity gives equation (B.14). Dealer optimisation gives $q = -\ell/\Gamma$, and market clearing gives equation (B.15). At the initial zero-premium allocation, the numerator of the investor's demand is $K_0 n_0$. Independent additional risk changes its numerator by $-\varrho r_j y_j V$ and its denominator by $\varrho r_j^2 V$. Substituting into $n - \ell/\Gamma = n_0$ gives equation (7). Also $n - n_0 = \ell/\Gamma$. Thus a negative premium is accompanied by a fall in the unhedged position and a rise in $H = A - n$.

For a positive trade response B^A, the credibility-adjusted trade prediction, the dollar depreciates when the portfolio contribution outweighs it,

$$\frac{\Lambda_{US}}{1 - \nu w} \frac{\Gamma \varrho r_U (y_U + n_0 r_U) V(D)}{\Gamma + K_0 + \varrho r_U^2 V(D)} > B^A(D),$$

where Λ_{US} is the U.S. impact multiplier of Appendix B.1 and the factor $1/(1 - \nu w)$ puts the portfolio contribution in dollar-index units. This is a restriction on tariff exposure, existing positions and dealer capacity. It links the variance generated by continuation to the credibility-adjusted mean. The two terms are evaluated for the same policy experiment.

Many currencies. Let the dollar be quoted against currencies $i = 1, \ldots, N$, with b_i its log value against currency i. A premium ℓ_U that originates in U.S. policy risk is a premium on dollar claims, priced in the dollar book of equation (B.15) regardless of the counterparty currency, so it moves the dollar against every floating currency by the same amount, $\Lambda_{US}\ell_U/(1 - \nu w)$ in the units of the dollar index. A global risk factor with premium ℓ_G moves the dollar against currency i according to the exposure θ_i of b_i, the covariance of the currency's price of a dollar with global risk, for which the currency's estimated response to the VIX provides a proxy. The trade block adds the JS response to the tariff τ_i imposed on partner i. To first order,

$$\Delta b_i = \frac{\Lambda_{US}\ell_U}{1 - \nu w} + \theta_i \ell_G + \kappa_{x,i} \Delta\tau_i + \Delta b^T, \tag{B.16}$$

where Δb^T collects the trade responses common to all partners, such as the offset to the U.S. import tariff. A U.S.-specific shock ($\ell_G = 0$, $\ell_U < 0$) lowers the dollar against all currencies by an amount unrelated to θ_i, with a dispersion that follows the announced tariffs. A global shock ($\ell_G > 0$) produces a dispersion that follows θ_i. These are the two cross-sectional regressions of Table 5, Panel B.

B.4 Official intervention and index aggregation

With the rule $I = -kB$ and equation (8), $B = U/(1+\beta k)$. Defining $t = k/(1+\beta k)$ and $g = \gamma + c\beta$, the equilibrium is

$$I = -tU, \quad B = (1-\beta t)U, \quad S = (c-gt)U + \chi, \quad F = -\gamma t U + \chi, \qquad 0 \le t \le c/g. \tag{B.17}$$

Because $t \le c/g < 1/\beta$, $1 - \beta t > 0$ and B has the sign of U. A rule $I = -kS$ would attenuate the direct trade response χ as well; the rule $I = -kB$ leaves it intact.

Let N_{US} and N_{CN} be effective currency appreciations, with China's weight ν in the U.S. effective index. The common index identities are

$$N_{US} = \nu S + (1-\nu)B, \quad N_{CN} = -F, \quad S = cB + F.$$

It follows that

$$B = \frac{N_{US} + \nu N_{CN}}{1 - \nu w}, \qquad F = -N_{CN}. \tag{B.18}$$

We apply the JS small-open-economy real block to each effective rate and aggregate through these identities. Regional baskets are alternative measurements of the same currency prices, rather than extra independent disturbances.

For the two segmented currency books, write

$$M_j = \frac{\Gamma_j K_j}{\Gamma_j + K_j}, \qquad L_j = \Lambda_j M_j, \quad j \in \{US, CN\}.$$

The units of official positions incorporate the relevant market-size conversion. A purchase I of dollar claims and sale I of renminbi claims produces $\Delta N_{US} = L_{US} I$ and $\Delta N_{CN} = -L_{CN} I$. Equation (B.18) gives

$$\beta = \frac{L_{US} - \nu L_{CN}}{1 - \nu w}, \qquad \gamma = L_{CN}. \tag{B.19}$$

The maintained positive-dollar-impact region is $L_{US} \ge \nu L_{CN}$. When a given purchase moves the dollar's effective value by no more than the renminbi's, $L_{US} \le L_{CN}$, as in a dollar market at least as deep as the renminbi market, and

$$0 \le \frac{\beta}{\gamma} = \frac{L_{US}/L_{CN} - \nu}{1 - \nu w} \le \frac{1-\nu}{1-\nu w}. \tag{B.20}$$

With $\nu = 0.109$ and $w = 0.320$ the bound is 0.92. A private dollar-demand shift contributes $L_{US} d/(1-\nu w)$ to U. More general correlated dealer books replace the diagonal costs with a positive-definite risk-cost matrix and produce the corresponding matrix of price impacts. The propositions use the scalar case, which prices private hedging and official transactions in a common clearing system.

B.5 The anchor choice

To link the policy weights to the tariff distribution, let Y^P be the country's non-currency income exposure and

$$\omega(D) = \omega_0 + \xi[-\operatorname{Cov}(R, Y^P \mid D)]_+, \qquad \kappa_s(D) = \kappa_0 + \kappa_1 \mathcal{O}\, m(D), \tag{B.21}$$

where $\xi, \kappa_1 \geq 0$ and $\mathcal{O}$ indicates that the anchor issuer is the trade opponent. The first term values the insurance supplied by the dollar; the second scales expected escalation costs with the credible trade exposure. Additional U.S. risk with $r_U y_U^P > 0$ lowers the insurance term and, for the opponent, raises κ_s through the expected lasting tariff. The constants κ_0 and κ_1 are institutional: the manipulator designation of August 2019 and the exchange-rate commitments of January 2020 made bilateral depreciation an explicit object of U.S. trade policy, so the model allows them to rise at those dates and predicts a weaker anchor response before them. The strategic term penalises depreciation only, and the parity is set daily, so the commitment is to a rule over an interval of news rather than to a fixed formula; its observable implication is that the loading in force before an announcement governs the transmission of that announcement (Section 6).

The loss function excludes two alternative motives. A competitiveness motive adds a reward for effective depreciation when tariffs rise, which reduces intervention against dollar appreciation and loosens the anchor in that case. A financial-stability motive adds a penalty on bilateral depreciation during capital outflows; it acts like κ_s on the dollar-appreciation side but scales with outflow pressure, and therefore binds most in years of dollar strength. The strategic term binds whenever bilateral depreciation is an object of the trade dispute, whatever the dollar's direction over the year.

Proof of Proposition 3. Substitute equation (B.17) into the loss. The terms in F^2 and I^2 give the strictly convex quadratic $\frac{1}{2}(\gamma^2 + \zeta)t^2\mathbb{E}[U^2]$ plus terms linear in t, and S^2 and $(S_+)^2$ are convex in t, so the loss is strictly convex. Because $(S_+)^2$ is continuously differentiable in t, with derivative $-2gUS_+$, differentiation under the expectation gives equation (11). Where U places no mass at the point at which S changes sign, the second derivative is

$$\mathcal{J}''(t) = (\gamma^2 + \zeta)\mathbb{E}[U^2] + \omega g^2 \mathbb{E}[U^2] + \kappa_s g^2 \mathbb{E}[U^2 \mathbf{1}\{S > 0\}] > 0. \tag{B.22}$$

The solution is unique, with the boundary chosen when the first-order condition has no interior root. For $\chi = 0$ and $0 \leq t \leq c/g$, S has the sign of U. The objective becomes

$$\mathcal{J}(t) = \frac{\mathbb{E}[U^2]}{2}\left[A_0 t^2 + (\omega + \kappa_s \pi_+)(c - gt)^2\right], \qquad A_0 = \gamma^2 + \zeta.$$

The first-order condition yields $t^* = \lambda g c/(A_0 + \lambda g^2)$. The loading relative to the observed dollar is $a = (c - gt)/(1 - \beta t)$. Substitution gives

$$a^* = \frac{cA_0}{A_0 + \lambda\gamma g},$$

using $g - c\beta = \gamma$, so that $c/a^* - 1 = \lambda\gamma g/A_0 = \lambda\mathcal{C}$, which is equation (12). Its derivatives are

$$\frac{\partial a^*}{\partial \omega} = -\frac{cA_0\gamma g}{(A_0 + \lambda\gamma g)^2} < 0, \qquad \frac{\partial a^*}{\partial \kappa_s} = \pi_+ \frac{\partial a^*}{\partial \omega}.$$

For unchanged π_+ and financial coefficients, a^* falls precisely when $\Delta\omega + \pi_+\Delta\kappa_s > 0$. If the distribution of dollar shocks also changes, the relevant change is $\Delta(\omega + \kappa_s\pi_+)$, together with any change in the coefficients of equation (12).

For general χ, implicit differentiation of equation (11) gives

$$\frac{\partial t^*}{\partial \omega} = \frac{g\mathbb{E}[US]}{\mathcal{J}''(t^*)}, \qquad \frac{\partial t^*}{\partial \kappa_s} = \frac{g\mathbb{E}[US_+]}{\mathcal{J}''(t^*)}.$$

These formulas state the corresponding conditions with direct trade news and asymmetric depreciation risk. Because the rule is chosen before the event, its realised response is to U; the coefficient on the realised direct component χ remains one. The strategic cost concerns the full bilateral change, including χ, through the ex ante choice in equation (11).

Corollary B.1 (A sign-contingent rule)**.** *Let the authority choose separate responses, t_+ to $U > 0$ and t_- to $U < 0$, with $\chi = 0$ and the financial coefficients of equation (B.17) on both sides. The optimal responses are $t^*_\pm = \lambda_\pm gc/(A_0 + \lambda_\pm g^2)$ with $\lambda_+ = \omega + \kappa_s$ and $\lambda_- = \omega$, and the loadings on dollar appreciation and dollar depreciation satisfy*

$$\frac{c}{a^*_+} - 1 = (\omega + \kappa_s)\mathcal{C}, \qquad \frac{c}{a^*_-} - 1 = \omega\mathcal{C}. \tag{B.23}$$

Hence $a^*_+ \le a^*_-$, with equality only when $\kappa_s = 0$, and for $\omega > 0$, $(c/a^*_+ - 1)/(c/a^*_- - 1) - 1 = \kappa_s/\omega$.

Proof of Corollary B.1. With $\chi = 0$ and $0 \le t_\pm \le c/g$, S has the sign of U, so $S_+ = S$ when $U > 0$ and $S_+ = 0$ otherwise. The loss separates into

$$\mathcal{J} = \frac{\mathbb{E}[U^2\mathbf{1}\{U > 0\}]}{2}\left[A_0 t_+^2 + (\omega + \kappa_s)(c - gt_+)^2\right] + \frac{\mathbb{E}[U^2\mathbf{1}\{U < 0\}]}{2}\left[A_0 t_-^2 + \omega(c - gt_-)^2\right],$$

and each bracket is minimised separately. The substitution used in the proof of Proposition 3 gives $a^*_\pm = cA_0/(A_0 + \lambda_\pm \gamma g)$, which is equation (B.23). Because $1 - \beta t_\pm > 0$, the observed dollar change has the sign of U, so the two loadings are those on days when the dollar appreciates and depreciates against the basket. The linear rule of Proposition 3 imposes $t_+ = t_-$; its first-order condition then weights the two brackets by π_+ and $1 - \pi_+$ and gives $\lambda = \omega + \kappa_s \pi_+$.

Corollary B.2 (Credibility and the anchor decision)**.** *Let $\mathcal{C}(D) = \gamma(D)g(D)/(\gamma(D)^2 + \zeta)$. In the pure-dollar case, with a positive insurance term in equation (B.21), fixed payoff exposures and fixed π_+,*

$$\frac{da^*}{dD} = -\frac{c\{\mathcal{C}[\pi_+\kappa_1\mathcal{O}m'(D) - \xi r_U y_U^P V'(D)] + \lambda\mathcal{C}'(D)\}}{(1 + \lambda\mathcal{C})^2}. \tag{B.24}$$

The credible mean, policy variance and asset-market price impact jointly determine the direction of anchor adjustment. At a fixed $\mathcal{C}$, the opponent tightens its anchor when $\pi_+\kappa_1 m'(D) > \xi r_U y_U^P V'(D)$; an economy with $\mathcal{O} = 0$ loosens it when $r_U y_U^P > 0$.

The same $p(D)$ thus determines both the expected trade exposure that raises the cost of bilateral depreciation and the risk that changes the dollar's insurance value.

Proof of Corollary B.2. Write $a^* = c/(1 + \lambda\mathcal{C})$. While the insurance term is positive, equation (B.21) implies $\omega' = -\xi r_U y_U^P V'$ and $\kappa_s' = \kappa_1 \mathcal{O} m'$. Holding π_+ fixed gives $\lambda' = \pi_+\kappa_1\mathcal{O}m' - \xi r_U y_U^P V'$. The chain rule yields equation (B.24). Financial coefficients may change through $K_j(D)$ in equation (B.19); their contribution is retained in $\mathcal{C}'$. The final two statements set this contribution to zero and compare the two values of $\mathcal{O}$.

B.6 Where U.S.-specific dollar risk is absorbed

Proposition B.2 (Strategic anchoring transfers dollar risk)**.** *For a U.S.-specific portfolio shock with pre-intervention dollar response $U_U < 0$ and no direct renminbi trade component,*

$$B_U = (1 - \beta t)U_U < 0, \qquad S_U = (c - gt)U_U \leq 0, \qquad F_U = -\gamma t U_U \geq 0. \tag{B.25}$$

A tighter rule reduces the bilateral renminbi appreciation and raises its effective depreciation, and with $\beta > 0$ it also reduces the dollar's depreciation. Equivalently,

$$F_U = \frac{(a - c)U_U}{1 + (\beta/\gamma)(c - a)}, \qquad \frac{\partial F_U}{\partial a} = \frac{U_U}{[1 + (\beta/\gamma)(c - a)]^2} < 0. \tag{B.26}$$

A tighter bilateral anchor thus comes with a larger effective renminbi response to adverse dollar risk. The result survives the stabilising feedback of official purchases to the dollar: the strategic channel changes where the risk is borne even when it reduces the original dollar movement. The feedback coefficient β/γ, bounded by equation (B.20), governs the size of that offset; the direction of the transfer does not depend on it.

Proof of Proposition B.2. Equation (B.17) with $\chi = 0$ gives equation (B.25). Its derivative $\partial F_U/\partial t = -\gamma U_U$ is positive for $U_U < 0$. Also $\partial B_U/\partial t = -\beta U_U \geq 0$, so the dollar decline becomes smaller. Eliminating $k = (c - a)/\gamma$ from equation (9) gives equation (B.26). Differentiation produces the displayed derivative. Thus the policy amplifies the renminbi's effective response even while it dampens the dollar shock that triggers intervention.

B.7 The JS special case and the testable implications

The model reduces to JS when continuation is certain, announcements are unanticipated, the additional portfolio premia vanish, intervention is zero and the loading is that of a free float: $\delta = 0$, $\ell_j = 0$, $I = 0$ and $a = c$. The zero-net-wealth convention of Appendix B.1 fixes the ownership of initial claims, and with additional positions and transfers set to zero and $\Gamma_j \to 0$ the premia and official purchases vanish. Equations (B.1)–(B.4) with $\ell = 0$ are then the JS equilibrium conditions for each economy, the index map (B.18) applies them to each effective rate, and every price response of Section 3 reduces to the permanent JS prediction.

Outside that case the model yields six implications. (T1) The own-component response per unit of announced exposure falls with the announced exposure for a given tariff-imposing country, and the implemented share of announced tariffs is a fraction comparable to the price response; Section 4 tests both. (T2) With a brief temporary tariff path, forward premia do not respond to tariff news while the spot response is a fraction of the permanent prediction; a temporary tariff path of three months or more implies forward responses of several basis points (Section 4). (T3) When the risk is U.S.-specific, the dollar's response contains a common component across currencies unrelated to safe-haven status, while relative movements follow the trade block (Section 5). (T4) Under the rule $I = -kB$, the own-component coefficient does not depend on the loading, and the transmission of the dollar's movement into the bilateral rate equals the loading in force before the news (Sections 4 and 6). (T5) The opponent's loading falls when a credible tariff raises the strategic cost; the fall is visible in the policy instrument and is larger on dollar appreciation when the rule can be sign-contingent, while other economies' loadings follow the insurance term (Section 6). (T6) A lower loading transfers U.S.-specific dollar weakness into effective renminbi depreciation (Section 7).

C Additional results

This appendix reports the announced exposure split at fixed cutoffs, the sensitivity of the own-component coefficient, the forward curve and dollar rates, the cross-currency loadings and break dates, the dollar's covariance with the VIX, and the main estimates under other values of the assigned JS parameters.

Table C1: Announced exposure split at fixed cutoffs

Cutoff (bp)	Period	Small doses	Large doses	Equality p	News days
50	2018–19	1.04 (0.29)	0.19 (0.06)	0.285	12 / 15
50	2025–26	1.28 (0.06)	0.16 (0.08)	0.630	4 / 10
100	2018–19	0.42 (0.20)	0.15 (0.04)	0.316	21 / 6
100	2025–26	0.28 (0.13)	0.15 (0.08)	0.459	10 / 4

Notes: Two-day responses from equation (2), splitting announcements at the stated absolute value of χ^{JS} across both countries. CR1 standard errors are in parentheses; equality tests use bootstrap p-values. Counts list small and large exposures. In 2018–19, nine of twelve small-exposure days at the 50 bp cutoff concern Chinese measures; Table 3 therefore tests size within U.S. measures.

Table C2: The own-component coefficient under alternative choices

Specification	2018–19	2025–26
Baseline: two days	0.20 (0.06)	0.18 (0.08)
Five-day window	0.37 (0.10)	0.19 (0.12)
Prior 250-day loading in place of the period loading	0.20 (0.06)	0.18 (0.09)
First announcements only (repeated doses set to zero)	0.19 (0.05)	0.16 (0.08)
Dollar basket without Asian currencies	0.24 (0.07)	0.20 (0.08)
Excluding windows across holiday closures	0.22 (0.08)	0.15 (0.07)
Second round restricted to 2025	0.20 (0.06)	0.16 (0.07)
Placebo: own change over the five days before the news	0.15 (0.09)	0.01 (0.06)
news days dropped by the holiday rule	06 May 2019; 08 Oct 2021; 05 Feb 2025; 24 Feb 2026	

Notes: Each row changes one specification choice in equation (2). The holiday restriction excludes windows spanning more than six calendar days between the common trading days before and after the news. The placebo relates the preceding five-day own-component change to the announcement-day prediction, controlling for news within that earlier window. CR1 standard errors are in parentheses.

Table C3: Forward premia and dollar rates on bilateral news

	2018–19 (bp)	p	2025 (bp)	p
One-month forward premium	0.37 (0.71)	0.626	0.64 (0.58)	0.384
Three-month forward premium	0.55 (1.40)	0.730	1.72 (1.26)	0.306
Six-month forward premium	0.22 (2.02)	0.933	1.95 (1.74)	0.337
One-year forward premium	−0.47 (2.42)	0.878	1.61 (2.09)	0.543
Three-month dollar OIS	−0.03 (0.34)	0.947	−0.27 (0.18)	0.255
Three-month Treasury bill minus OIS	−0.09 (0.52)	0.868	0.32 (0.50)	0.659

Notes: Two-day responses to signed bilateral news (bp), with period intercepts. CNH forward premia use Bloomberg closing quotes and equal $10^4 \log(F_H/S)$. Dollar OIS rates reference federal funds through January 2020 and the Secured Overnight Financing Rate in 2024–25; Treasury bill rates are from H.15. CR1 standard errors are in parentheses; bootstrap p-values test a zero response. The second-round sample is 2025.

Table C4: Twenty currencies: loadings, breaks and the reciprocal tariffs

Currency	2023 loading	2026 loading	Change	Best single break	Reciprocal tariff (pp)	Safe-haven beta	Dollar, 2–3 April 2025 (bp)
NOK	1.574 (0.114)	0.914 (0.112)	−0.660	Jul 2024	15	13.9	−75
AUD	1.382 (0.082)	1.050 (0.081)	−0.332	Jun 2024	10	11.7	−82
CNY	0.508 (0.039)	0.185 (0.030)	−0.323	Nov 2024	34	–	–
TWD	0.452 (0.041)	0.254 (0.056)	−0.198	Jul 2017	32	3.0	24
DKK	1.024 (0.046)	0.851 (0.040)	−0.172	Nov 2022	20	2.7	−166
GBP	0.998 (0.062)	0.856 (0.056)	−0.142	Dec 2018	10	6.7	−107
EUR	0.856 (0.052)	0.743 (0.043)	−0.113	Nov 2022	20	2.6	−168
NZD	1.370 (0.086)	1.276 (0.078)	−0.094	May 2018	10	7.1	−128
CAD	0.577 (0.045)	0.485 (0.061)	−0.092	Nov 2022	0	6.7	−175
SGD	0.668 (0.022)	0.606 (0.026)	−0.062	Jul 2021	10	3.9	−55
SEK	1.458 (0.088)	1.401 (0.087)	−0.057	Dec 2017	20	8.5	−153
THB	1.011 (0.070)	0.955 (0.119)	−0.056	Nov 2022	36	4.3	0
INR	0.165 (0.039)	0.165 (0.082)	0.000	Mar 2022	26	2.8	−9
JPY	0.788 (0.116)	0.798 (0.137)	0.010	Jun 2022	24	−4.1	−266
MYR	0.365 (0.051)	0.459 (0.095)	0.095	Nov 2022	24	3.3	−24
ZAR	1.591 (0.116)	1.779 (0.145)	0.188	Jun 2022	30	11.9	−12
CHF	0.887 (0.061)	1.098 (0.073)	0.211	Mar 2025	31	−0.9	−263
BRL	0.814 (0.129)	1.068 (0.116)	0.254	Aug 2022	10	10.3	−174
MXN	0.577 (0.124)	1.031 (0.095)	0.454	Aug 2022	0	14.3	−272
KRW	0.589 (0.108)	1.224 (0.137)	0.635	May 2025	25	7.6	−79
Correlation of the change with the reciprocal tariff, 19 currencies other than CNY: 0.08							

Notes: Loadings use days without tariff news and exclude each currency from its reference basket where applicable. HC1 standard errors are in parentheses; 2026 covers January–August. The single break minimises residual variation with at least 120 days on either side. Tariff rates follow Executive Order 14257, Annex I. Safe-haven sensitivity is the 2017–2024 weekly exchange-rate coefficient on VIX changes (bp per index point). The final column gives dollar changes between noon on 2 and 3 April 2025 (bp). The final three columns provide the data for Figure 3.

Table C5: Break dates in the renminbi's loading

Segment	Loading	No-news days	95% interval for the break
A. Bai–Perron segments (BIC selects the number of breaks)			
04 Jan 2017 – 12 Nov 2024	0.458 (0.019)	1787	–
13 Nov 2024 – 31 Aug 2026	0.189 (0.024)	380	23 Oct 2024 – 04 Mar 2025
B. Loading in the 250 no-news days before and after a date			
Date	Before / after	Equality p	Event
05 Aug 2019	0.601 / 0.381	0.004	U.S. designates China a currency manipulator
15 Jan 2020	0.469 / 0.454	0.861	Phase One agreement signed
05 Nov 2024	0.336 / 0.245	0.103	U.S. presidential election
25 Nov 2024	0.374 / 0.181	<0.001	First tariff threat of the second round
02 Apr 2025	0.358 / 0.179	<0.001	Reciprocal tariffs announced

Notes: Panel A reports Bai–Perron break estimates for the regression on days without tariff news, with at least 120 days per segment and up to five breaks. The Bayesian information criterion selects the number of breaks. Panel B compares loadings over the 250 days without tariff news before and after each date. HC1 standard errors are in parentheses; equality tests use HC1 p-values.

Table C6: Dollar–VIX covariance by year

Year	Daily	First five-day alignment	Minimum	Maximum
2018	3.20	2.75	2.28	8.02
2019	0.63	4.37	1.54	4.37
2020	0.72	1.51	1.51	7.95
2021	1.34	6.08	1.38	6.08
2022	7.42	11.03	10.24	15.38
2023	4.72	16.00	12.27	19.16
2024	0.92	−1.96	−1.96	4.99
2025	1.52	4.09	−4.02	4.74
2026	8.46	17.51	16.35	20.59

Notes: Coefficients from regressions of dollar changes against the basket (bp) on VIX changes (index points). Five-day estimates use non-overlapping common trading days and report the first starting-day alignment and the range across all five possible alignments.

Table C7: The main estimates under other values of the assigned JS parameters

Case	κ_m; κ_x (China)	Own-component slope			D_0	Priced / implemented share	
		2018–19	2025–26	Pooled	(bp)	2018–19	to Dec 2025
JS calibration	0.286; 1.000	0.20 (0.06)	0.18 (0.08)	0.19 (0.05)	47.5	0.24 / 0.26	0.20 / 0.20
$\epsilon_x = 2$	0.429; 1.000	0.18 (0.05)	0.10 (0.05)	0.13 (0.05)	42.8	0.23 / 0.26	0.10 / 0.15
$\epsilon_x = 5$	0.171; 1.000	0.24 (0.07)	0.49 (0.18)	0.28 (0.06)	52.7	0.27 / 0.27	0.51 / 0.31
$\epsilon_m = 2$	0.438; 0.787	0.26 (0.07)	0.22 (0.10)	0.24 (0.06)	51.6	0.31 / 0.26	0.24 / 0.20
ω_H lower by 0.05	0.273; 1.000	0.20 (0.06)	0.18 (0.08)	0.19 (0.05)	47.4	0.24 / 0.26	0.20 / 0.20
ω_H higher by 0.05	0.298; 1.000	0.20 (0.06)	0.18 (0.08)	0.19 (0.05)	47.6	0.24 / 0.26	0.20 / 0.20
2017 trade shares in 2025–26	0.286; 1.000	0.20 (0.06)	0.09 (0.04)	0.12 (0.04)	43.0	0.24 / 0.26	0.09 / 0.19

Notes: Each row changes one JS calibration assumption and re-estimates the two-day own-component coefficients, D_0, and the priced and implemented shares for 2018–19 and announcements through December 2025. CR1 standard errors are in parentheses. The export elasticity also governs substitution across import sources; home bias changes equally in both countries. The final row uses 2017 trade shares for the second round.

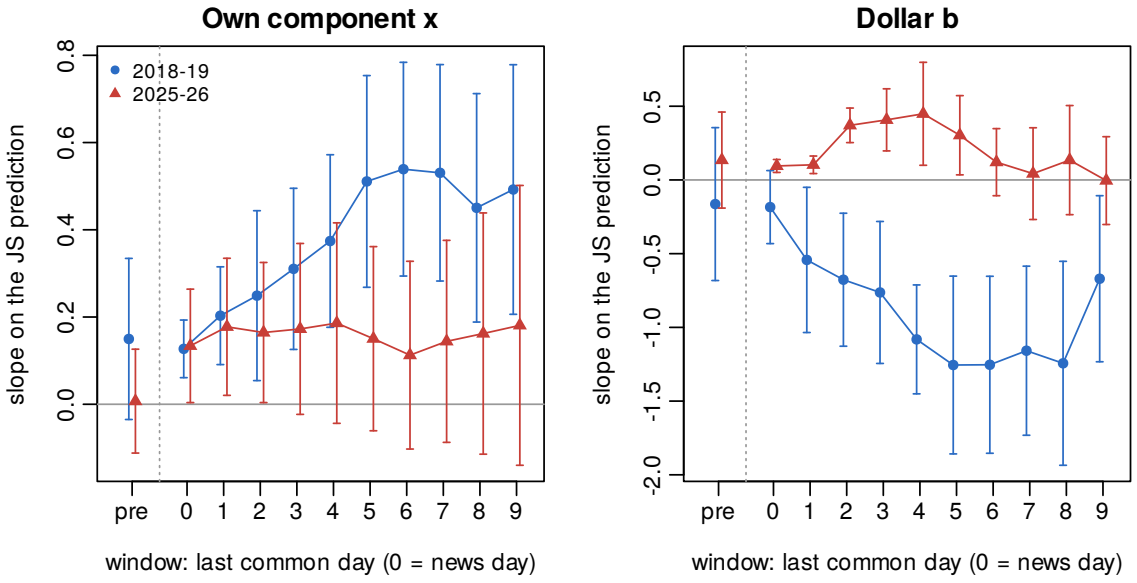


Figure C1: The calibration coefficient by window length. *Notes:* Coefficients from equation (2) for windows ending $h = 0, \ldots, 9$ common trading days after the news, with corresponding lead and lag controls. "Pre" denotes the five-day placebo before the announcement. Bars show 95% confidence intervals based on CR1 standard errors.

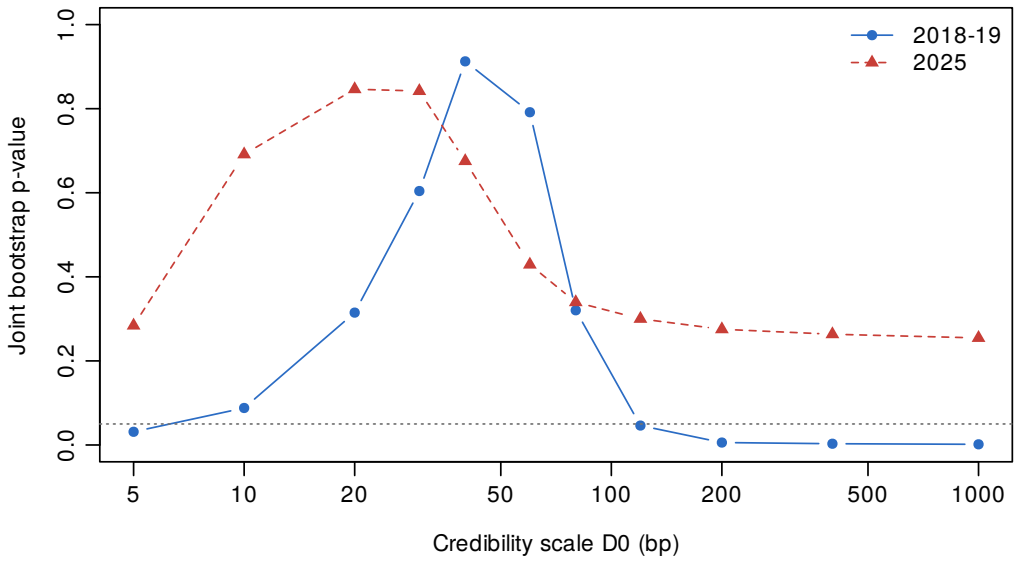


Figure C2: Joint spot and forward tests over the credibility scale. *Notes:* Bootstrap p-values for the joint spot–forward test in Table 4 across values of D_0, with a mean temporary-tariff duration of 14 days. The horizontal line marks 0.05. The second-round sample is 2025.

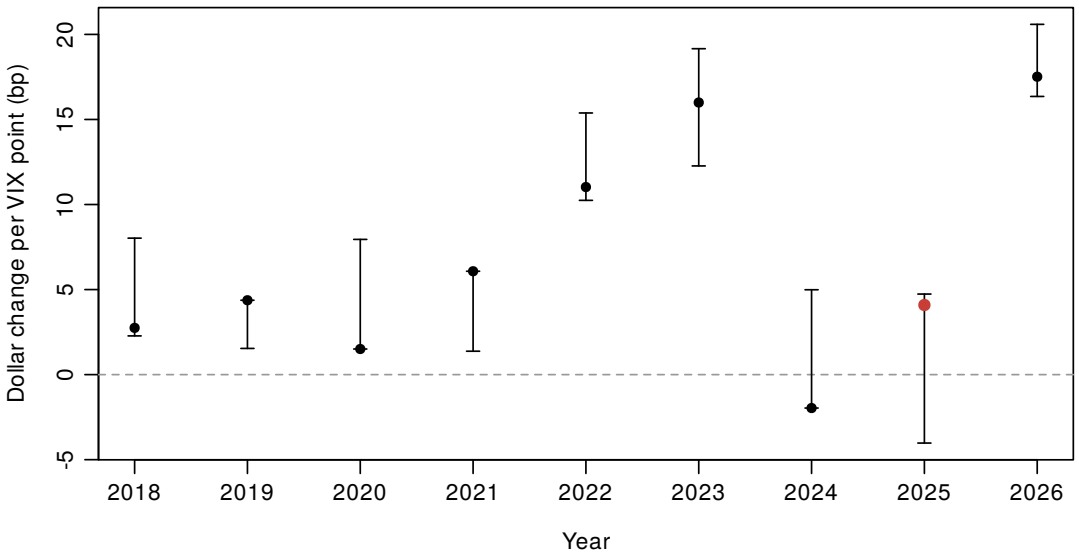


Figure C3: The dollar's covariance with the VIX. *Notes:* Five-day coefficients from Table C6: first starting-day alignment (points) and range across the five alignments (bars). The 2025 estimate is shown in red.